It is Taxing to be Coherent*

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Abstract

The presence of an occasionally binding constraint due to the zero lower bound (ZLB) in New Keynesian models generally gives rise to multiple equilibria under active monetary policy. To restore uniqueness in the model with an active Taylor Rule, we consider appropriate simple fiscal policy instruments. Without relaxing the assumptions of full information and rational expectations, we show that appropriate fiscal targeting rules ensure that New Keynesian models subject to the ZLB possess a unique solution.

Keywords: fiscal policy, incompleteness, incoherency, rational expectations, zero lower bound

JEL Codes: C62, E4, E61, E62, E63

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1 Introduction

The canonical New Keynesian (NK) model with an occasionally binding constraints arising from the zero lower bound (ZLB), with full information rational expectations (FIRE) and an active Taylor Rule (TR), exhibits multiple minimum state variable (MSV) solutions. Thus, it is termed as "incomplete". Furthermore, when subjected to significant shocks, an MSV solution may not exist, rendering it "incoherent". These crucial findings were demonstrated in seminal works by Ascari and Mavroeidis (2022) (AM) in a stochastic environment with rational expectations and Holden (2022) under perfect foresight.

In absence of the ZLB, a standard linearised NK model satisfies the coherency and completeness (CC) conditions, defined as the existence of a unique MSV solution. For the standard NK model, this is illustrated as the intersection of the aggregate demand (AD) and aggregate supply (AS) curves. However, the presence of the occasionally binding constraint in the ZLB regime introduces a piecewise linear structure to the model, violating the CC conditions. When the economy is not subject to fundamental shocks (Mertens and Ravn 2014), this violation occurs due to the steeper unconstrained segment of the AD curve compared to the AS curve, caused by an interest rate rule that adheres to the Taylor principle. In the presence of fundamental shocks, the violation of CC conditions is driven by the lack of an active policy response to these shocks, failing to resolve uncertainty in the economy.

This paper, maintaining FIRE, demonstrates that simple Ricardian fiscal policy (FP) can ensure coherency and completeness even when nominal interest rates follow active Taylor-type rules. The results hold robustly for shocks of any size and support. The key insight is: if FP is sufficiently persistent and reactive to fluctuations, then it guarantees a unique MSV solution while also satisfying local determinacy conditions. We show

two properties to achieve this. First, at the ZLB, FP provides an active stabilisation mechanism when monetary policy is constrained, pinning down an equilibrium path for the economy. Second, uncertainty-driven equilibria are ruled out when FP is set according to a countercyclical rule that is sufficiently persistent.

To emphasise the importance of this paper and the relevance of coherency and completeness, it is worth contextualising and clarifying the main contributions of AM and the literature. While considering shocks in models subject to the ZLB, the literature has often employed stylised approaches, such as assuming a single structural shock – occurring once – that brings the nominal interest rate to the ZLB or imposing tight assumptions such as a fixed known duration for the shock. In contrast, AM consider the case of multiple structural shocks and their serial occurrences in a dynamic forward-looking model with FIRE. Although this paper abstracts from multiple structural shocks, it examines the recurring structural shock scenario. Building upon the work of Gourieroux, Laffont, and Monfort (1980) (GLM), AM derive two main results using a linearised system of equations and endogenous regime switching. First, they establish that achieving coherency in NK models with the ZLB is a nontrivial challenge. When the TR coefficient on inflation exceeds unity or even when monetary policy is optimal under discretion, coherency may not be satisfied. Additionally, AM derive conditions that impose restrictions on the support of stochastic shocks, which are necessary to ensure model coherency. However, these support restrictions are cumbersome, as they depend on model structural parameters and – when considering a backwards looking model – past realisations of state variables. Second, even if support restrictions are applied to the structural shocks to satisfy coherency, the model may still fail to satisfy completeness by admitting multiple MSV solutions. AM demonstrate the existence of multiple MSV solutions, potentially reaching up to 2^k

solutions, where k represents the number of discrete states the shocks can assume, as in Rouwenhorst (1995).

The scope of this concern goes beyond what the ZLB literature has traditionally examined, such as sunspot shocks or belief-driven fluctuations between multiple steady states; despite more recent papers focusing on the existence of equilibria – the problem of model coherency.[1] However, to the best of our knowledge, there are no general conditions to ensure model coherency in the macroeconomics DSGE literature, though a recent number of papers provide sufficient conditions for the existence of MSV equilibria in NK models (Eggertsson 2011; Christiano, Eichenbaum, and Johannsen 2018; Nakata 2018; Nakata and Schmidt 2019).

As highlighted by Ascari, Mavroeidis, and McClung (2022) in subsequent research, the multiplicity of MSV solutions arises from the interaction between rational expectations and the inherent non-linear nature of the ZLB constraint. While they focus on relaxing the assumptions of rational expectations, this paper maintains the FIRE framework and proposes alternative mechanisms, specifically emphasising the role of simple Ricardian FP, to address the issues identified by AM. Our objective is to provide qualitative results on resolving the problem of MSV solution multiplicity rather than offering precise quantitative analysis or specific policy recommendations. Additionally, despite this paper speaking to fiscal and monetary policy interactions (Galí, López-Salido, and Vallés 2007; Davig and Leeper 2011; Eggertsson and Krugman 2012; Billi and Walsh 2022; Hills and Nakata 2018), it avoids discussing the potency of fiscal policy and fiscal multipliers at the ZLB.

The rest of the paper is structured as follows. Section 2 provides an overview of the CC

1. See, for example, Eggertsson and Woodford (2003), Guerrieri and Iacoviello (2015), Kulish, Morley, and Robinson (2017), Aruoba, Cuba-Borda, and Schorfheide (2018), Aruoba et al. (2021), and Angeletos and Lian (2022)

conditions in the context of a NK model subject to the ZLB and includes a description of the methodology that we use to verify if the model satisfies the CC conditions. In Section 3, using a purely forward-looking reference NK model with a ZLB constraint, we show how Ricardian FP restores coherency and completeness. Section 4 provides assessment of CC conditions for an NK model with FP with an endogenous state variable. Section 5 concludes the paper.

2 A New Keynesian Model with the ZLB

2.1 A Canonical Model

This section provides an overview of the canonical NK model subject to the ZLB, and conditions required to adhere to the CC conditions. In other words, we illustrate the conditions that the model must satisfy to have a unique equilibrium in the presence of occasionally binding constraints.

Consider the canonical NK model as set out in, for example, Galí (2015). The model in its log-linearised form with the ZLB can be written in three equations, the dynamic IS equation (DISE), New Keynesian Phillips Curve (NKPC), and the TR:[2]

$$\text{DISE: } \hat{y}_t = \mathbb{E}_t \hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}) + \varepsilon_t, \tag{1a}$$

$$\text{NKPC: } \hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa \hat{y}_t, \tag{1b}$$

$$\text{TR: } \hat{i}_t = \max\left\{-\mu, \phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t\right\}, \tag{1c}$$

and where ε_t is a demand shock. Furthermore, $\hat{y}_t$ is the output gap, $\hat{\pi}_t$ is the net inflation

2. To keep the analysis simple, we omit cost-push shocks in the NKPC and monetary policy shocks in the TR.

rate, and $\hat{i}_t$ is the net nominal interest rate.[3] The parameters of interest in the model are: σ, the coefficient of relative risk aversion; β, the representative household's subjective discount factor; κ, the slope of the NKPC; $\mu = \log(r\pi^*)$, the ZLB of the nominal interest rate in deviation from the steady state, where $r = 1/\beta$ is the steady state gross real interest rate and π^* is the gross inflation target of the monetary authority; ϕ_y, the monetary authority's response parameter to output fluctuations; and ϕ_π, the monetary authority's responsiveness to inflation.

2.2 Verifying the Coherency and Completeness Conditions of Linear Models

As stated in AM, many solution methods of log-linearised models which feature the ZLB, such as Eggertsson and Woodford (2003), Guerrieri and Iacoviello (2015), Kulish, Morley, and Robinson (2017), Eggertsson and Singh (2019), and Holden (2022), can be verified for coherency in a simple manner. Let $\boldsymbol{Y}_t$ be a $n \times 1$ vector of endogenous variables, $\boldsymbol{X}_t$ be a $n_x \times 1$ vector of exogenous state variables, and $s_t \in \{0, 1\}$ be an indicator variable that is equal to 1 when some inequality constraint is slack and 0 otherwise.[4] This then allows us to write state-space models, and thus DSGE models, in the form:

$$\begin{aligned} \mathbf{0} &= \boldsymbol{A}_{s_t}\boldsymbol{Y}_t + \boldsymbol{B}_{s_t}\boldsymbol{Y}_{t+1|t} + \boldsymbol{C}_{s_t}\boldsymbol{X}_t + \boldsymbol{D}_{s_t}\boldsymbol{X}_{t+1|t} + \boldsymbol{H}_{s_t}\boldsymbol{Y}_{t-1}, \\ s_t &= \mathbb{1}\{\boldsymbol{a}^\top\boldsymbol{Y}_t + \boldsymbol{b}^\top\boldsymbol{Y}_{t+1|t} + \boldsymbol{c}^\top\boldsymbol{X}_t + \boldsymbol{d}^\top\boldsymbol{X}_{t+1|t} + \boldsymbol{h}^\top\boldsymbol{Y}_{t-1} > 0\}, \end{aligned} \tag{2}$$

3. Hatted variables denote a variable in terms of log deviations from steady state. In other words, for any generic variable, say, X, we have:

$$\hat{x}_t = \ln X_t - \ln \bar{X} \approx \frac{X_t - \bar{X}}{\bar{X}},$$

where $\bar{X}$ is the value of X_t in the non-stochastic steady state.

4. Additionally, let $\boldsymbol{\Omega}_t$ denote the information set, thus allowing us to write: $\boldsymbol{Y}_{t+1|t} = \mathbb{E}_t[\boldsymbol{Y}_{t+1}|\boldsymbol{\Omega}_t]$ and $\boldsymbol{X}_{t+1|t} = \mathbb{E}_t[\boldsymbol{X}_{t+1}|\boldsymbol{\Omega}_t]$.

where $\boldsymbol{A}_{s_t}$, $\boldsymbol{B}_{s_t}$, $\boldsymbol{C}_{s_t}$, $\boldsymbol{D}_{s_t}$, and $\boldsymbol{H}_{s_t}$ are coefficient matrices with dimensions $n \times n$, $n \times n$, $n \times n_x$, $n \times n_x$, and $n \times n$, respectively; $\boldsymbol{a}$, $\boldsymbol{b}$, $\boldsymbol{c}$, $\boldsymbol{d}$, and $\boldsymbol{h}$ are coefficient vectors, and $\mathbb{1}\{\cdot\}$ is an indicator function that is equal to 1 if its argument holds true and 0 otherwise. The key contribution of the paper by AM is that it analyses (2) with rational expectations and Markovian shocks with discrete support. This was as opposed to GLM which analysed the coherency of a system like (2) when $\boldsymbol{B}_{s_t} = \boldsymbol{H}_{s_t} = \boldsymbol{O}$ and $\boldsymbol{b} = \boldsymbol{h} = \mathbf{0}$. In other words, with no endogenous state variables and no expectations on future realisations of the endogenous variables. If the model features endogenous state variables, in other words, $\boldsymbol{H}_{s_t} \neq \boldsymbol{O}, \boldsymbol{h} \neq \mathbf{0}$, the canonical form (2) is not piecewise linear and, thus, the standard approach presented by GLM does not apply. For this section and the next, we abstract from the models that feature endogenous state variables.[5]

Coherency requires that for the system (2) there exist some function $f(\cdot)$ such that an MSV solution can be represented as $\boldsymbol{Y}_t = f(\boldsymbol{X}_t)$. Assume that the exogenous states $\boldsymbol{X}_t$ are k-state stationary first-order Markov processes with transition kernel $\boldsymbol{K}$. Stack the possible states of $\boldsymbol{X}_t$ for states $i = 1, .., k$ into a $n_x \times k$ matrix $\mathbf{X}$. Then, let $\boldsymbol{e}_i$ denote the i-th column of the $k \times k$ identity matrix $\boldsymbol{I}_k$, such that $\mathbf{X}\boldsymbol{e}_i$, the i-th column of $\mathbf{X}$, is the i-th state of $\boldsymbol{X}_t$.[6] Then define $\mathbf{Y}$ as an $n \times k$ matrix whose i-th column, $\mathbf{Y}\boldsymbol{e}_i$, corresponds to $\boldsymbol{X}_t = \mathbf{X}\boldsymbol{e}_i$ along a MSV solution. Thus, along a MSV solution we have:

$$\mathbb{E}[\boldsymbol{Y}_{t+1}|\boldsymbol{Y}_t = \mathbf{Y}\boldsymbol{e}_i] = \mathbb{E}_t[\boldsymbol{Y}_{t+1}|\boldsymbol{X}_t = \mathbf{X}\boldsymbol{e}_i] = \mathbf{Y}\boldsymbol{K}^\top \boldsymbol{e}_i.$$

5. We revisit CC conditions for models with endogenous states in Section 4, where we study the baseline NK model with persistent FP rules.

6. The elements of the transition kernel $\boldsymbol{K}$ are $\boldsymbol{K}_{ij} = \Pr(\boldsymbol{X}_{t+1} = \mathbf{X}\boldsymbol{e}_j|\boldsymbol{X}_t = \mathbf{X}\boldsymbol{e}_i)$ and hence, $\mathbb{E}_t[\boldsymbol{X}_{t+1}|\boldsymbol{X}_t = \mathbf{X}\boldsymbol{e}_i] = \mathbf{X}\boldsymbol{K}^\top \boldsymbol{e}_i$.

Substituting this into (2), yields:

$$\begin{aligned} \mathbf{0} &= (\boldsymbol{A}_{s_i}\mathbf{Y} + \boldsymbol{B}_{s_i}\mathbf{Y}\boldsymbol{K}^\top + \boldsymbol{C}_{s_i}\mathbf{X} + \boldsymbol{D}_{s_i}\mathbf{X}\boldsymbol{K}^\top)\boldsymbol{e}_i, \\ s_i &= \mathbb{1}\{(\boldsymbol{a}^\top\mathbf{Y} + \boldsymbol{b}^\top\mathbf{Y}\boldsymbol{K}^\top + \boldsymbol{c}^\top\mathbf{X} + \boldsymbol{d}^\top\mathbf{X}\boldsymbol{K}^\top)\boldsymbol{e}_i > 0\},\ i = 1, ..., k. \end{aligned} \tag{3}$$

This system of equations relates $\mathbf{Y}$ to $\mathbf{X}$, and can be expressed as $F(\mathbf{Y}) = \lambda(\mathbf{X})$, where $\lambda(\cdot)$ is some function of $\mathbf{X}$, and $F(\cdot)$ is a piecewise linear continuous function of $\mathbf{Y}$. If $J \subseteq \{1, ..., k\}$, then the piecewise linear function $F(\mathbf{Y})$ can then be expressed as:

$$F(\mathbf{Y}) = \sum_J \boldsymbol{\mathcal{A}}_J \boldsymbol{S}_{\boldsymbol{\mathcal{C}}_J} \mathrm{vec}(\mathbf{Y}), \tag{4}$$

where $\boldsymbol{\mathcal{C}}_J = \{\mathbf{Y} : \mathbf{Y} \in \mathbb{R}^{n\times k}, s_i = \mathbb{1}_{\{i\in J\}}\}$ is given by a configuration of regimes over the k states given by J, $\boldsymbol{S}_{\boldsymbol{\mathcal{C}}_J}$ is a $nk \times nk$ matrix of indicator elements,[7] and $\mathrm{vec}(\cdot)$ is the vector operator function.[8] In words: $\boldsymbol{\mathcal{A}}_J$ and $\boldsymbol{\mathcal{C}}_J$ are such that if $F(\mathbf{Y})$ in (4) is invertible, then the linear system is coherent and complete. Put another way, there exists a unique MSV solution, as stipulated in GLM,[9] if all the determinants of $\boldsymbol{\mathcal{A}}_J$, $J \subseteq \{1, ..., k\}$ share the same sign. Failure of this requirement implies that the model is generally incoherent and/or incomplete.

7. Note that when $\mathbf{Y}_{n\times k}$ is vectorised, and if $k = 2$, the first n elements correspond to state 1 and the last n elements correspond to state 2. Thus, in essence, the elements of $\boldsymbol{S}_{\boldsymbol{\mathcal{C}}_J}$ map the entries of $\mathcal{A}_J$ for the k states to the vectorised set of endogenous variables in $\mathbf{Y}$.

8. The transformation of (3) into (4) is generally non-trivial (in which the expressions of $\boldsymbol{\mathcal{A}}_J$ require Kronecker product operations) as it presents a Sylvester equation in $\mathbf{Y}$. See, for example, Kolmogorov and Fomin (1957). However, there are two exceptions that allow straightforward computation of the $\boldsymbol{\mathcal{A}}_J$: $n = 1$ and $n = k > 1$. We make use of this simplifying assumption both in this example and the analytical derivation in Section 2.2.1.

9. This follows Theorem 1 of GLM:

Theorem 1 (GLM). *Suppose that the mapping $F(\cdot)$ defined in (4) is continuous. A necessary and sufficient condition for $F(\cdot)$ to be invertible is that all the determinants $\det \mathcal{A}_J, J \subseteq \{1, ..., k\}$ have the same sign.*

Example: Simple Fisherian Model. To demonstrate the methods of AM and GLM, consider the simple model taken from Section 2 of Aruoba, Cuba-Borda, and Schorfheide (2018), which consists of the Euler equation,

$$1 = \mathbb{E}_t \left[M_{t,t+1} \frac{1+i_t}{\pi_{t+1}} \right], \tag{5}$$

and a TR that only targets inflation,

$$1 + i_t = \max \left\{ 1, r\pi^* \left(\frac{\pi_t}{\pi^*} \right)^{\phi_\pi} \right\}, \quad \phi_\pi > 1, \tag{6}$$

where $M_{t,t+1}$ is the stochastic discount factor and the steady value of the gross real interest rate is given by $r = 1/M = (1+i)/\pi$. The law of motion of $M_{t,t+1}$ is given by a 2-state Markov process with a transitory state $r^{-1}\exp(-r^L) > r^{-1}$ and an absorbing state r^{-1} with transition probabilities p and q, respectively.[10]

Combine the two equations above and log-linearise about the non-stochastic steady state to get

$$\mathbb{E}_t \hat{\pi}_{t+1} = \mathbb{E}_t \hat{M}_{t,t+1} + \max \left\{ -\mu, \phi_\pi \hat{\pi}_t \right\},$$

which can be cast in canonical form (3) as follows:

$$0 = \left(-\phi_\pi \begin{pmatrix} \hat{\pi}_t^1 & \hat{\pi}_t^2 \end{pmatrix} + \begin{pmatrix} \hat{\pi}_{t+1}^1 & \hat{\pi}_{t+1}^2 \end{pmatrix} \boldsymbol{K}^\top + \begin{pmatrix} -1 & 0 \end{pmatrix} \begin{pmatrix} \frac{\exp(-r^L)}{r} & 0 \\ 1 & 1 \end{pmatrix} \boldsymbol{K}^\top \right) \boldsymbol{e}_i,$$

10. $r^L < 0$ is a simple negative real interest rate shock, representing a temporary liquidity trap.

if $s_t = \mathbb{1}\{\phi_\pi \hat{\pi}_t + \mu > 0\}$. However, if $s_t = \mathbb{1}\{\phi_\pi \hat{\pi}_t + \mu < 0\}$ then we have

$$0 = \left(\begin{pmatrix} \hat{\pi}^1_{t+1} & \hat{\pi}^2_{t+1} \end{pmatrix} \boldsymbol{K}^\top + \begin{pmatrix} 0 & \mu \end{pmatrix} \begin{pmatrix} \frac{\exp(-r^L)}{r} & 0 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 0 \end{pmatrix} \begin{pmatrix} \frac{\exp(-r^L)}{r} & 0 \\ 1 & 1 \end{pmatrix} \boldsymbol{K}^\top \right) \boldsymbol{e}_i,$$

where the transition matrix $\boldsymbol{K}$ is

$$\boldsymbol{K} = \begin{pmatrix} p & 1-p \\ 1-q & q \end{pmatrix}, \tag{7}$$

and where $\mathbf{Y}_t = \hat{\pi}_t$ and $\boldsymbol{X}_{t+1} = \begin{pmatrix} \hat{M}_{t,t+1} & 1 \end{pmatrix}^\top$. The coefficient matrices[11] are given as $\boldsymbol{A}_0 = 0$, $\boldsymbol{A}_1 = -\phi_\pi$, $\boldsymbol{B}_0 = \boldsymbol{B}_1 = 1$, $\boldsymbol{C}_0 = \begin{pmatrix} 0 & \mu \end{pmatrix}$, $\boldsymbol{C}_1 = \begin{pmatrix} 0 & 0 \end{pmatrix}$, and $\boldsymbol{D}_0 = \boldsymbol{D}_1 = \begin{pmatrix} -1 & 0 \end{pmatrix}$. The coefficient vectors are given as $\boldsymbol{a} = \phi_\pi$, $\boldsymbol{b} = 0$, $\boldsymbol{c} = \begin{pmatrix} 0 & \mu \end{pmatrix}^\top$, and $\boldsymbol{d} = \begin{pmatrix} 0 & 0 \end{pmatrix}^\top$.

The coefficient matrices $\boldsymbol{\mathcal{A}}_J$ in (4) are given by

$$\begin{aligned} \boldsymbol{\mathcal{A}}_{J_1} &= \boldsymbol{A}_1 \boldsymbol{I}_2 + \boldsymbol{B}_1 \boldsymbol{K}, \quad J_1 = \{1, 2\}, \\ \boldsymbol{\mathcal{A}}_{J_2} &= \boldsymbol{e}_1 \boldsymbol{e}_1^\top \boldsymbol{\mathcal{A}}_{J_4} + \boldsymbol{e}_2 \boldsymbol{e}_2^\top \boldsymbol{\mathcal{A}}_{J_1}, \quad J_2 = \{2\}, \\ \boldsymbol{\mathcal{A}}_{J_3} &= \boldsymbol{e}_2 \boldsymbol{e}_2^\top \boldsymbol{\mathcal{A}}_{J_4} + \boldsymbol{e}_1 \boldsymbol{e}_1^\top \boldsymbol{\mathcal{A}}_{J_1}, \quad J_3 = \{1\}, \\ \boldsymbol{\mathcal{A}}_{J_4} &= \boldsymbol{A}_0 \boldsymbol{I}_2 + \boldsymbol{B}_0 \boldsymbol{K}, \quad J_4 = \emptyset. \end{aligned} \tag{8}$$

As $J \subseteq \{1, ..., k\}$ and $k = 2$ – and J contains all configurations of combinations of the k states – we can think of the above equations as transitions between positive and negative states. Specifically, positive (PIR) and zero interest rate (ZIR) states: $\boldsymbol{\mathcal{A}}_{J_1}$ and

11. In this example, since $n = 1$, some of the coefficient matrices are scalars.

$J_1 = \{1, 2\}$ is associated with being in the PIR state and remaining in the PIR state, $\boldsymbol{\mathcal{A}}_{J_2}$ and $J_2 = \{2\}$ is associated with being the ZIR state and transitioning to the PIR state, $\boldsymbol{\mathcal{A}}_{J_3}$ and $J_3 = \{1\}$ is associated with being in the PIR state and transitioning to the ZIR state, and $\boldsymbol{\mathcal{A}}_{J_4}$ and $J_4 = \{\emptyset\}$ is associated with being in the ZIR state and remaining in the ZIR state.

Substituting the coefficient matrices into (8), the determinants of the $\boldsymbol{\mathcal{A}}_J$ are:

$$\begin{aligned}
|\boldsymbol{\mathcal{A}}_{J_1}| &= (\phi_\pi - 1)(1 - p - q + \phi_\pi), \\
|\boldsymbol{\mathcal{A}}_{J_2}| &= p(1 - \phi_\pi) + q - 1, \\
|\boldsymbol{\mathcal{A}}_{J_3}| &= p - 1 + q(1 - \phi_\pi), \\
|\boldsymbol{\mathcal{A}}_{J_4}| &= p + q - 1.
\end{aligned} \tag{9}$$

Since $\phi_\pi > 0$ (satisfaction of the Taylor principle), and $0 \leq p, q \leq 1$, it is straightforward to see that $|\boldsymbol{\mathcal{A}}_{J_1}| > 0$ and $|\boldsymbol{\mathcal{A}}_{J_2}|, |\boldsymbol{\mathcal{A}}_{J_3}| < 0$, and so this is a violation of the CC conditions according to Theorem 1 of GLM.

2.2.1 Coherency and Completeness of the Canonical New Keynesian Model

Below, we provide a sketch of the insight of AM as applied to the canonical NK model, (1), but for simplicity $\phi_y = 0$. Then assume as before that $k = 2$ and the transition matrix is given by (7). The relevant coefficient matrices of the canonical form (3) are given by

$$\boldsymbol{A}_0 = \begin{pmatrix} 1 & -\kappa \\ 0 & 1 \end{pmatrix}, \quad \boldsymbol{A}_1 = \begin{pmatrix} 1 & -\kappa \\ \frac{\phi_\pi}{\sigma} & 1 \end{pmatrix}, \quad \boldsymbol{B}_0 = \boldsymbol{B}_1 = \begin{pmatrix} -\beta & 0 \\ -\frac{1}{\sigma} & -1 \end{pmatrix}.$$

Observe that, in the special case where $p = q = 1$, the determinants of $\boldsymbol{\mathcal{A}}_{J_1}$ and $\boldsymbol{\mathcal{A}}_{J_4}$ are:

$$|\boldsymbol{\mathcal{A}}_{J_1}| = \begin{vmatrix} 1-\beta & -\kappa \\ \frac{\phi_\pi - 1}{\sigma} & 0 \end{vmatrix} = \frac{\kappa(\phi_\pi - 1)}{\sigma} > 0, \tag{10}$$

$$|\boldsymbol{\mathcal{A}}_{J_4}| = \begin{vmatrix} 1-\beta & -\kappa \\ -\frac{1}{\sigma} & 0 \end{vmatrix} = -\frac{\kappa}{\sigma} < 0. \tag{11}$$

Thus, we observe that with an active TR ($\phi_\pi > 1$) the function $F(\mathbf{Y})$ is not invertible and, hence, the model is generally incomplete. We provide detailed exposition in Appendix A and a general proof in Appendix A.1.[12]

Additionally, in Appendix A.1 we show an analytical derivation of the CC conditions. Denoting ϕ_π^* as

$$\phi_\pi^* = p + q - 1 - \frac{\sigma}{\kappa}(1 - \beta(p + q - 1))(2 - p - q),$$

then the model satisfies the CC conditions when

$$\phi_\pi < \phi_\pi^*, \quad \text{if } \phi_\pi^* > 0, \tag{12a}$$

$$\phi_\pi < 1, \quad \text{if } \phi_\pi^* < 0. \tag{12b}$$

We can graphically represent the CC conditions for the canonical NK model by first considering the absorbing state of the model for when $\varepsilon_t = 0$. In the absorbing state we have $\hat{\pi}_t = \hat{\pi}_{t+1} = \hat{\pi}$ and $\hat{y}_t = \hat{y}_{t+1} = \hat{y}$. Hence, the NKPC can be written as the following

12. We refer readers to AM for a more thorough discussion.

aggregate supply (AS) relation:

$$\hat{\pi} = \frac{\kappa}{1-\beta}\hat{y} \quad AS. \tag{13}$$

Meanwhile, the DISE can be written and rearranged to give a piecewise aggregate demand (AD) relation:

$$\hat{\pi} = \begin{cases} \frac{\kappa\phi_\pi}{1-\beta}\hat{y} & AD^{TR}, \\ -\mu & AD^{ZLB}. \end{cases} \tag{14}$$

Clearly, the model admits two absorbing states: a PIR equilibrium, $\{\hat{\pi}, \hat{y}, \hat{i}\} = \{0, 0, 0\}$, and a ZIR equilibrium, $\{\hat{\pi}, \hat{y}, \hat{i}\} = \{-\mu, -\frac{\mu(1-\beta)}{\kappa}, -\mu\}$, which we can graphically see by plotting (13) and (14) as in Figure 1.

The left plot of Figure 1 shows the incompleteness problem when the NK model features an active TR. Absent of any shocks, the model implies two equilibria as the slope of AD^{TR} is steeper than that of AS. By contrast, when $\phi_\pi < 1$, as in the right plot of Figure 1, a unique equilibrium exists as the AD and AS curves intersect once. However, as is well studied in the literature, a passive TR leads to issues with local model dynamics (Blanchard and Kahn 1980) (BK).

Now consider the transitory state for when $\varepsilon_t = \frac{p}{\sigma}\hat{r}^T$. For simplicity, we assume that the shock is transitory and occurs once, in other words $q = 1$. As the model is completely forward looking, the economy remains in the transitory state for some indefinite period of time after which it jumps to an absorbing state – either a PIR or ZIR equilibrium.

PIR absorbing state. At time t the economy is in a transitory state. With probability p the economy remains in the transitory state $(\hat{y}^T, \hat{\pi}^T)$; with complimentary probability $1-p$ the economy moves to the PIR absorbing state. Thus, the AS and AD relations

can be written as

$$\hat{\pi}^T = \frac{\kappa}{1-p\beta}\hat{y}^T \quad AS, \tag{15a}$$

$$\pi^T = \begin{cases} \frac{\sigma(1-p)}{p-\phi_\pi}\hat{y}^T - \frac{p}{p-\phi_\pi}\hat{r}^T & AD^{TR} \text{ for } \hat{\pi}^T \geq -\frac{\mu}{\phi_\pi}, \\ \frac{\sigma(1-p)}{p}\hat{y}^T - \frac{\mu}{p} - \hat{r}^T & AD^{ZLB} \text{ for } \hat{\pi}^T \leq -\frac{\mu}{\phi_\pi}. \end{cases} \tag{15b}$$

ZIR absorbing state. Here we repeat the above exercise but for when the absorbing state is a ZIR equilibrium. As before, at time t the economy is in a transitory state, and with probability p it remains in the transitory state, and with probability $1-p$ it transitions to the ZIR absorbing state. As previously mentioned, the absorbing state here now differs in value from the PIR case, and as such, the AS and AD relations can be written as:

$$\hat{\pi}^T = \frac{\kappa}{1-p\beta}\hat{y}^T - \frac{\beta(1-p)}{1-p\beta}\mu \quad AS, \tag{16a}$$

$$\hat{\pi}^T = \begin{cases} \frac{\sigma(1-p)}{p-\phi_\pi}\hat{y}^T + \frac{1-p}{p-\phi_\pi}\left[\frac{(1-\beta)}{\kappa}+1\right]\mu - \frac{p}{p-\phi_\pi}\hat{r}^T & AD^{TR} \text{ for } \hat{\pi}^T \geq -\frac{\mu}{\phi_\pi}, \\ \frac{\sigma(1-p)}{p}\hat{y}^T + \frac{1-p}{p}\left[\frac{(1-\beta)\sigma}{\kappa}+1\right]\mu - \frac{\mu}{p} - \hat{r}^T & AD^{ZLB} \text{ for } \hat{\pi}^T \leq -\frac{\mu}{\phi_\pi}, \end{cases} \tag{16b}$$

respectively. To complete the description of this simple example, we define θ as the ratio of the slopes of the AD^{ZLB} and AS relations:

$$\theta = \frac{\sigma(1-p)(1-p\beta)}{p\kappa}. \tag{17}$$

Figure 2 then plots the AS and AD when monetary policy adheres to the Taylor principle ($\phi_\pi > 1$) for either a PIR or ZIR absorbing state when the economy is subject to the shock term ε_t. The plots on the left hand side of Figure 2 are for the case of $\theta < 1$, i.e.,

when AD is flatter relative to AS. The plots on the right in Figure 2 are for the case where $\theta > 1$, i.e., when AD is more steep than AS. The different values for θ correspond to different values of p. Namely, the plots shown in Subfigure 2b with $\theta < 1$ are for higher values of p than those generated for the case where $\theta > 1$. Additionally, the higher value of p corresponds to a higher probability that the model remains in a transitory state each period.

As discussed by AM, and shown in Figure 2, for the case where $\theta > 1$, the only support restriction necessary for a MSV solution to exist in the absorbing state is $(r\pi^*)^{-1} \leq 1$. But for when $\theta \leq 1$, the necessary support restriction becomes:

$$\frac{1}{r\pi^*} = 1 \text{ and } -\hat{r}^L \leq \log(r\pi^*)\left(\frac{\phi_\pi - p}{\phi_\pi p} + \frac{\theta}{\phi_\pi}\right). \tag{18}$$

To put it simply, these support restrictions ensure that a negative shock to AD does not lead it shifting too far to the left or above of AS, as shown in $AD_1^{TR,ZLB}$ of Subfigure 2b.

Derivations and further explanation can be found in Appendix A. However, we encourage readers to refer to AM for detailed discussion of the CC conditions of the NK model when subject to the ZLB, including examples as applied to Aruoba, Cuba-Borda, and Schorfheide (2018) and optimal monetary policy under discretion from Nakata (2018) and Nakata and Schmidt (2019).

We highlight the fact that non-uniqueness of equilibria in the baseline NK model is driven by exogenous uncertainty captured by p. We proceed with analysing how simple Ricardian fiscal policy can counteract the effects of uncertainty and restore uniqueness.

3 Fiscal Policy and the Coherency and Completeness Conditions

We augment the baseline NK model with a simple FP setup following Woodford (2011). The model is otherwise standard, and derivation is given in Appendix B. In what follows, we show that under simple fiscal feedback rules, the model can generate a unique equilibrium in the presence of the ZLB under certain restrictions on FP. The model is described by the DISE, NKPC, and TR given by:

$$\hat{x}_t = \mathbb{E}_t \hat{x}_{t+1} - \frac{c}{\sigma}\left(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1} - \hat{r}_t^n\right), \tag{19a}$$

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_y \hat{x}_t - \kappa_g \widehat{GR}_t, \tag{19b}$$

$$\hat{i}_t = \max\{-\mu; \phi_\pi \hat{\pi}_t + \phi_y \hat{x}_t\}, \tag{19c}$$

and with the natural interest rate,

$$\hat{r}_t^n = \frac{\sigma \psi_{ya}}{c} \Delta \mathbb{E}_t \hat{a}_t - \mathbb{E}_t \Delta \hat{\tau}_{t+1}^c - \frac{\sigma g}{c} \mathbb{E}_t \Delta \widehat{GR}_{t+1} - \frac{\sigma}{c} \mathbb{E}_t \Delta \hat{z}_{t+1}, \tag{20}$$

where $c = C/Y$ is the steady-state consumption-output ratio, $g = G/Y$ is the steady-state government expenditure-output ratio, and κ_y and κ_g denote slopes of the NKPC and coefficient on government revenue, GR_t, respectively. Additionally, total factor productivity (TFP) is a_t, consumption taxes are τ_t^c, government expenditure is g_t, and z_t are household preference shocks.

Strict inflation targeting MP ($\phi_y = 0$). Start by assuming for analytical tractability that $\hat{a}_t = 0$, $\varepsilon_t = -\mathbb{E}_t \Delta \hat{z}_{t+1}$, and that MP only closes the inflation gap. Additionally, make

the assumption that the fiscal authority only levies lump-sum taxes to fund government expenditure, and it is unable to issue debt. Thus, $\tau_t^c = 0$, and $\hat{g}_t = \hat{\tau}_t = \widehat{GR}_t$.[13] Then (20) can be written compactly as

$$\hat{r}_t^n = -\frac{\sigma g}{c}\mathbb{E}_t \Delta \widehat{GR}_{t+1} - \frac{\sigma}{c}\varepsilon_t. \tag{21}$$

We close the model with a simple contemporaneous fiscal feedback rule of the form

$$\widehat{GR}_t = \psi_\pi \hat{\pi}_t + \psi_y \hat{x}_t, \tag{22}$$

where ψ_π and ψ_y denote the degree of reaction of taxes to deviations of inflation and the output gap, respectively. Tax income enters the model equations because the government operates a balanced budget. In this case, only lump-sum taxes are levied, so there is a one-to-one mapping of lump-sum taxes to government spending.

The model can be cast in canonical form as in (3), and we check if the model satisfies the CC conditions. As a reminder, the CC conditions are satisfied if the determinants of (8) have the same sign.

Proposition 1. *A baseline New Keynesian model with a simple fiscal rule described by (19) and (21)-(22), in which monetary policy only closes the inflation gap ($\phi_y = 0$) and the Taylor principle is satisfied ($\phi_\pi > 1$), fails to satisfy the coherency and completeness conditions.*

Rearranging the system of equations and obtaining the relevant matrices from the canonical form with relevant coefficients provided in Appendix B.2, CC conditions are

13. We also assume that $\tau_t^w = 0$. We will revisit this assumption later.

satisfied if and only if $sign|\boldsymbol{\mathcal{A}}_{J_1}| = sign|\boldsymbol{\mathcal{A}}_{J_4}|$. This is not the case since

$$
\begin{aligned}
|\boldsymbol{\mathcal{A}}_{J_1}| &= |\boldsymbol{A}_1 + \boldsymbol{B}_1\boldsymbol{I}_2| \\
&= \begin{vmatrix} \frac{c}{\sigma}(1-\phi_\pi) & 0 \\ \beta - 1 - \kappa_g\psi_\pi & \kappa_y - \kappa_g\psi_y \end{vmatrix} \\
&= \frac{c(1-\phi_\pi)(\kappa_y - \kappa_g\psi_y)}{\sigma}, \\
|\boldsymbol{\mathcal{A}}_{J_4}| &= \frac{c(\kappa_y - \kappa_g\psi_y)}{\sigma}.
\end{aligned} \tag{23}
$$

But (23) gives $|\boldsymbol{\mathcal{A}}_{J_1}| = |\boldsymbol{\mathcal{A}}_{J_4}|(1-\phi_\pi)$, which implies that under an active TR, $\phi_\pi > 1$, the simple NK model with FP rule (22) does not generally satisfy the CC conditions.

As in the baseline NK model without FP, this can be seen graphically in the case of the absorbing state ($\varepsilon_t = 0$) or the transitory state with a PIR absorbing state. We illustrate the case of the absorbing state with $\psi_y = 0$ by rearranging the system of equations into AS and AD schedules, both for when the ZLB binds and does not bind:

$$
\hat{\pi} = \frac{\kappa_y}{1-\beta+\kappa_g\psi_\pi}\hat{x} \quad AS, \tag{24a}
$$

$$
\hat{\pi} = \begin{cases} \frac{\phi_\pi\kappa_y}{1-\beta+\kappa_g\psi_\pi}\hat{x} & AD^{TR}, \\ -\mu & AD^{ZLB}. \end{cases} \tag{24b}
$$

Notice that with the introduction of FP, both the AS and AD^{TR} are augmented and sensitive to two FP parameters: the NKPC coefficient of government expenditure, κ_g, and the FP rule reaction parameter to inflation, ψ_π. Furthermore, notice that the slopes of AS and AD^{TR} are dependent on the relative sizes of these parameters. Namely, if $\psi_\pi > (1-\beta)/\kappa_g$, then AS and AD are diagrammatically similar to the case of the baseline

NK model with no FP (Figure 1), with two solutions – one PIR and one ZIR. This is illustrated in the left diagram of Figure 3. However, if the condition $\psi_\pi < (1-\beta)/\kappa_g$ holds, then both AS and AD^{TR} become downward sloping, and if AD^{TR} is steeper than AS then only one unique solution remains – the PIR equilibrium. This case is illustrated in the right diagram of Figure 3.

But is the condition $\psi_\pi < (1-\beta)/\kappa_g$ enough to ensure a unique solution once the economy is subject to shocks? No. To see this diagrammatically, as done in Section 2.2.1, we consider the economy when it starts in the transitory state and is subject to shocks with probability p in each period, and with probability $1-p$ the economy transitions to its PIR absorbing state. AS and AD can then be written as follows:

$$\hat{\pi}^T = \frac{\kappa_y}{1-p\beta+\kappa_g\psi_\pi}\hat{x}^T \quad AS, \tag{25a}$$

$$\hat{\pi}^T = \begin{cases} \frac{\sigma(1-p)}{pc+(1-p)\sigma g\psi_\pi - c\phi_\pi}\hat{x}^T - \frac{p}{cp+(1-p)g\sigma\psi_\pi - c\phi_\pi}\hat{r}^T & AD^{TR}, \\ \frac{\sigma(1-p)}{pc+(1-p)\sigma g\psi_\pi}\hat{x}^T - \frac{c}{pc+g\sigma\psi_\pi(1-p)}\mu - \frac{p}{cp+(1-p)g\sigma\psi_\pi}\hat{r}^T & AD^{ZLB}. \end{cases} \tag{25b}$$

Figure 4 plots AS and AD from (25), where we can clearly see that regardless of the relative size of κ_g and ψ_π, the CC conditions are not satisfied and thus, in general, no solution exists or that two solutions exist. This result holds true even when the condition $\psi_\pi < (1-\beta)/\kappa_g$ holds, which is the condition needed to ensure a unique absorbing state.

3.1 Coherent and Complete Simple Fiscal Policy Rules

In this section, we explore simple FP rules to ensure the satisfaction of the CC conditions of the canonical NK model subject to the ZLB. In what follows, we provide analytical derivations and numerical results, computed with parameter values as shown in Table 1,

which are standard in the NK literature.

MP targets the inflation and the output gap. Extending the analysis from Section 3, now consider the case where the monetary authority targets both the inflation and output gap ($\phi_y > 0$).

As before, rearrange the model for the absorbing state where $\varepsilon_t = 0$, and express the model in two equations, AS and AD:

$$\hat{\pi} = \Theta \hat{x} \quad AS, \tag{26a}$$

$$\hat{\pi} = \begin{cases} (\phi_\pi \Theta + \phi_y)\, \hat{x} & AD^{TR}, \\ -\mu & AD^{ZLB}, \end{cases} \tag{26b}$$

where

$$\Theta = \frac{\kappa_y - \kappa_g \psi_y}{1 - \beta + \kappa_g \psi_\pi}.$$

Note carefully that the slopes of AS and AD in (26) are potentially ambiguous. In line with the calibration in Table 1 and with ψ_y, ψ_π being sufficiently large in absolute value, the following assumptions on parameter values are made:

$$\Theta < 0,$$

and

$$\phi_\pi \Theta + \phi_y > \Theta \implies \Theta(1 - \phi_\pi) < \phi_y.$$

One can see that the LHS of the above inequality is always positive, and that ϕ_y must be sufficiently large for the inequality to hold. This inequality highlights the importance

of MP targeting both inflation and the output gap. If MP follows a strict inflation targeting regime, the inequality would never be satisfied and, hence, no matter the fiscal policy stance (captured by Θ), the existence of multiple absorbing states is never ruled out. Hence, multiplicity of steady states in this case is only ruled out under a particular configuration of fiscal-monetary mix. In other words, should the slope of AS be positive, then, under conventional restrictions on MP parameters, the slope of AD^{TR} would also be positive and larger and steeper than that of AS, creating the two – PIR and ZIR – absorbing states. With these assumptions, AS and AD in (26) is plotted in Figure 5, and we have the following proposition:

Proposition 2. *A baseline New Keynesian model with a simple fiscal rule described by* (19a)-(19c) *and* (21)-(22)*, is coherent and complete if fiscal policy responds aggressively enough to inflation and the output gap.*

Next we show analytical results for when the economy is in the transitory state. As mentioned above, as the ZIR absorbing state is eliminated, we restrict attention to the existence of a unique PIR absorbing state. Assume that initially the economy is in a transitory state with $\varepsilon_t \neq 0$ and it will remain in this transitory state with probability p. The system can then be written as follows

$$\hat{\pi}^T = \frac{\kappa_y - \kappa_g \psi_y}{1 - p\beta + \kappa_g \psi_\pi} \hat{x}^T \quad AS, \tag{27a}$$

$$\hat{\pi}^T = \begin{cases} \frac{\sigma(1-p)(1-g\psi_y)+c\phi_y}{pc+(1-p)\sigma g\psi_\pi - c\phi_\pi}\hat{x}^T - \frac{p}{cp+(1-p)g\sigma\psi_\pi - c\phi_\pi}\hat{r}^T & AD^{TR}, \\ \frac{\sigma(1-p)(1-g\psi_y)}{pc+(1-p)\sigma g\psi_\pi}\hat{x}^T - \frac{c}{pc+g\sigma\psi_\pi(1-p)}\mu - \frac{p}{cp+(1-p)g\sigma\psi_\pi}\hat{r}^T & AD^{ZLB}. \end{cases} \tag{27b}$$

With ψ_π sufficiently large and ψ_y positive but not too large, AS is downward sloping and AD^{TR} is upward sloping. Since AD^{ZLB} is also upward-sloping ($p < 1$) or flat ($p = 1$),

there is a unique solution for any realisation of $\hat{r}^T$. The system (26) is illustrated in Figure 5.

Much like the case for the absorbing state, the downward sloping AS curve is central to the uniqueness result. This is predicated on: (i) direct influence of fiscal policy on aggregate supply, (ii) fiscal policy being procyclical, and (iii) the monetary authority also reacting to the output gap. Absent of either of the aforementioned points, the model would imply a non-unique solution and, thus, the policy stance presented above merely presents a special case that is not applicable to a more general class of models. First, absent of direct fiscal policy effects, the AS curve is always upward sloping as in a baseline NK model. This is true, for example, if there is no income effect on the household's labour supply decision due to preferences such as in Greenwood, Hercowitz, and Huffman (1988) (GHH), or if labour is supplied inelastically. In such a case, fiscal policy would not directly affect AS and thus, its slope would remain positive. Second, even if fiscal policy had direct effects on AS, it needs to react positively to deviations of inflation and output. If this were not to hold, AS would be upward sloping, which would generate multiple equilibria.

However, the result is robust to the calibration of the government expenditure share in output, g. To illustrate this, consider upper and lower bounds on ψ_y. The upper bound on ψ_y can be inferred from the restriction on AD^{ZLB} being upward sloping or flat, which is the case if and only if

$$1 - g\psi_y \geq 0 \implies \psi_y \leq \frac{1}{g}.$$

The lower bound on ψ_y can be inferred from the restriction on the slope of AS which

must be negative. This implies that

$$\kappa_y - \kappa_g \psi_y < 0 \implies \psi_y > \frac{\kappa_y}{\kappa_g}.$$

These conditions imply that the model is coherent and complete only if

$$\frac{\kappa_y}{\kappa_g} > \frac{1}{g} \implies -\varrho g^2 + (\varrho + 1)g - 1 > 0,$$

where $\varrho = \varphi/\sigma$. Thence, the relationship holds for $g < \min(1, 1/\varrho)$, which is always the case if the coefficient of relative risk aversion is greater than inverse-Frisch elasticity of labour supply, $\sigma > \varphi$.

Finally, to visually see the need for procyclical fiscal policy – and the upper and lower limits on ψ_y – we numerically compute regions for which CC conditions are satisfied in $\{\psi_\pi, \psi_y\}$ space in Figure 6.

Thus far, we have shown how fiscal policy, possessing direct supply- and demand-side effects, can ensure the satisfaction of CC conditions in a special case. In what follows, we analyse the effects of fiscal policy that has demand-side, but no direct supply-side effects.

Permanent fiscal policy changes. Here we explore whether a permanent change in fiscal policy stance can guarantee a unique MSV solution and satisfy the CC conditions. Focusing on permanent rather than inertial rules allows to study CC properties of the model in a simple analytical fashion. We generalise the analysis to inertial rules and allow for supply-side effects in Section 4.

The model in this section is the same as that described in (19). But in this section direct supply-side effects of fiscal policy are nonexistent. This can be achieved by either assuming that output taxes adjust such that the effect of government spending on ag-

gregate supply is offset or if the household's utility function takes the form of, say, GHH preferences whereby there is no income effect on the household's labour supply decision or that labour is supplied inelastically.

The NK-FP model used is:

$$\hat{x}_t = \mathbb{E}_t \hat{x}_{t+1} - \frac{c}{\sigma}(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1} - \hat{r}_t^n), \tag{28a}$$

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_y \hat{x}_t - \frac{\epsilon}{\Phi}\left(\frac{1}{\epsilon}\hat{\tau}_t^s + \frac{\sigma}{c}\hat{g}_t\right), \tag{28b}$$

$$\hat{i}_t = \max\{-\mu, \phi_\pi \hat{\pi}_t + \phi_y \hat{x}_t\}, \tag{28c}$$

$$\hat{g}_t = \widehat{GR}_t, \tag{28d}$$

where $\widehat{GR}_t$[14] is given by

$$\widehat{GR}_t = \frac{\tau}{Y}\hat{\tau}_t + \bar{\tau}^s(\hat{\tau}_t^s + \hat{x}_t + \hat{y}_t^n),$$

and the natural interest rate is given by

$$\hat{r}_t^n = -\frac{\sigma g}{c}\mathbb{E}_t \Delta \hat{g}_{t+1} - \frac{\sigma}{c}\mathbb{E}_t \Delta \hat{z}_{t+1}.$$

As stated, the key assumption used in this section is that government expenditure is offset by the production subsidy such that:

$$\hat{\tau}_t^s = -\frac{\sigma \epsilon}{g}\hat{g}_t. \tag{29}$$

With these assumptions, and re-using $\varepsilon_t = -\mathbb{E}_t \Delta \hat{z}_{t+1}$, the system of equations de-

14. The non-linear expression for tax revenue is:[15]

$$GR_t = Y_t \tau_t^s + \tau_t.$$

scribed in (28) can be written compactly as:

$$\hat{x}_t = \mathbb{E}_t \hat{x}_{t+1} - \frac{c}{\sigma}\left(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1} - \hat{r}_t^n\right), \tag{30a}$$

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_y \hat{x}_t, \tag{30b}$$

$$\hat{i}_t = \max\{-\mu; \phi_\pi \hat{\pi}_t + \phi_y \hat{x}_t\}, \tag{30c}$$

$$\hat{r}_t^n = -\frac{\sigma}{c}(g\mathbb{E}_t \Delta\widehat{GR}_{t+1} + \varepsilon_t), \tag{30d}$$

In this section we highlight two rules which are as follows. First, we assume that FP "activates" only when the economy hits the ZLB. Graphically, this can be seen as AD extending past the kink point of AD^{TR} and AD^{ZLB}, thereby ensuring the existence of a unique MSV solution. Second, we show how a more general FP rule can ensure the satisfaction of the CC conditions.

Finally, we show that the general conclusions with regard to coherency and completeness with FP span a rather large set of FP formulations. If the policymaker is able to affect aggregate demand without affecting aggregate supply and commit to a permanent change in policy, multiplicity of equilibria in the presence of the ZLB is ruled out.

The knife-edge FP rule. The first fiscal rule we inspect is what we term the "knife-edge" rule that replicates monetary policy at the ZLB and mirrors the approach in Correia et al. (2013). Assume that the government revenue growth rate, $\mathbb{E}_t \Delta\widehat{GR}_{t+1}$, responds to contemporaneous inflation and the output gap when the interest rate is at the ZLB:

$$\mathbb{E}_t \Delta\widehat{GR}_{t+1} = \mathbb{1}_{\{\hat{i}_t = -\mu\}}(\psi_\pi^{KE} \hat{\pi}_t + \psi_y^{KE} \hat{x}_t), \tag{31}$$

where ψ_π^{KE} and ψ_y^{KE} denote the coefficients of reaction to inflation and the output gap, respectively.

The presence of the FP instrument in the DISE allows the piecewise linear system to satisfy the CC conditions, despite the presence of the ZLB constraint on $\hat{i}_t$ and an active TR. The instrument $\mathbb{E}_t \Delta \widehat{GR}_{t+1}$ has the same effect in the NK model as the monetary policy instrument and, hence, it governs the linearity of the DISE (19a). The CC conditions are satisfied so long as:

$$\psi_\pi^{KE} = \frac{c}{g\sigma}\phi_\pi, \quad \psi_y^{KE} = \frac{c}{g\sigma}\phi_y, \tag{32}$$

which also allows (19c) to follow an active TR ($\phi_\pi > 1$). It is straightforward to see that since the model is now linear, it is generally coherent and complete. The rule is similar to Correia et al. (2013) who show that a set of tax instruments can replicate monetary policy when the interest rate is at the ZLB. This knife-edge rule also applies to models where monetary policy is strictly inflation targeting, whereby if $\phi_y = 0$ then $\psi_y = 0$. Thus, we have our first proposition:

Proposition 3. *A baseline New Keynesian model with fiscal policy that consists of government spending, lump-sum, and output taxes as defined in* (19)*, is generally coherent and complete when the sensitivity parameters of the fiscal instrument,* ψ_π^{KE} *and* ψ_y^{KE}*, allow fiscal policy to replicate monetary policy at the ZLB as described in the "knife-edge" fiscal rule Equation* (31)*.*

Coherency and completeness for the knife-edge rule is illustrated in Figure 7 for the special case where $\phi_y = \psi_y^{KE} = 0$. We plot AD and AS for both the absorbing (steady state) case where $\varepsilon_t = 0$ (Subfigure 7a) and the transitory state with a PIR absorbing

state (Subfigure 7b).

In the absence of active FP, the AD curve is illustrated, as before, with a piecewise red line, which may not intersect AS as shown with $AD^{ZLB,TR}$ in Subfigure 7a and AD_1 in Subfigure 7b. Once FP is activate at the ZLB, as in the "knife-edge" fiscal rule (31), it fully mimics monetary policy as if the latter was unconstrained. Thus, AD is a linear relation composed of the red AD^{TR} line and the purple AD^{KE} line. In other words, in the presence of active FP stemming from the knife edge rule, the model always has a unique solution.

3.1.1 FP targets the Inflation and Output Gap

Next we consider a more general rule that ensures the CC conditions of the NK model be satisfied. Assume that the fiscal authority contemporaneously targets inflation and the output gap using the following rule:

$$\mathbb{E}_t \Delta \widehat{GR}_{t+1} = \psi_\pi \hat{\pi}_t + \psi_y \hat{x}_t. \tag{33}$$

Then, we have the following proposition:

Proposition 4. *A baseline New Keynesian model with fiscal policy that consists of government spending, output taxes, and lump-sum taxes as defined in* (19)*, is generally coherent and complete when the reaction of fiscal policy to inflation and the output gap,* ψ_π *and* ψ_y*, respectively, as described in the fiscal rule Equation* (33)*, is sufficiently strong.*

Proof of Proposition 4 is provided in Appendix B.3. The region for which the model satisfies the CC conditions as a function of the fiscal authority's reaction parameters, ψ_π and ψ_y, are shown in Figure 8. Using our baseline calibration, we see that the model

generally satisfies the CC conditions in the negative orthant of ψ_π and ψ_y space, $\mathbb{R}^2_-$, and when ψ_π is sufficiently large. The intuition for this is that a strong enough reaction on the part of the fiscal authority to inflation and output deviations leads to a unique MSV solution by ensuring an intersection between AD and AS. We illustrate this for a simple case in Figure 10. Furthermore, we note that the degree of reaction on the part of the fiscal authority to the output gap is largely irrelevant as to whether or not the model satisfies the CC conditions.

As such, in order to clarify our intuition and understand the mechanism driving requirements of the CC conditions, we focus on a simplified version of (33) where the fiscal authority strictly targets inflation ($\psi_y = 0$) with the following rule:

$$\mathbb{E}_t \Delta \widehat{GR}_{t+1} = \psi_\pi \hat{\pi}_t. \tag{34}$$

We then plot the region for which the model satisfies the CC conditions, for parameters of ϕ_π and ψ_π as shown in the Figure 9. We emphasise two points: with the simple FP rule in Equation (34), the model satisfies the CC conditions despite the monetary authority adopting an active TR. This addresses one of the major concerns surrounding NK models subject to the ZLB as raised by AM. Secondly, in order to satisfy the CC conditions, the fiscal authority must respond strongly enough to inflation.

We term FP regimes in which an undesired ZIR equilibrium is ruled as "active FP". Consider the simple FP rule, Equation (34), and the absorbing state case with $\varepsilon_t = 0$. The AS curve takes a similar form as in Equation (13), and the AD curve is piecewise linear, giving us the following system:

$$\hat{\pi} = \frac{\kappa_y}{1-\beta}\hat{x} \quad AS, \tag{35a}$$

$$\hat{\pi} = \begin{cases} (\phi_\pi + \psi^*)\frac{\kappa_y}{1-\beta}\hat{x} & AD^{TR} \text{ for } \hat{\pi} > -\mu, \\ \psi^* \frac{\kappa_y}{1-\beta}\hat{x} - \mu & AD^{ZLB} \text{ for } \hat{\pi} < -\mu, \end{cases} \tag{35b}$$

where $\psi^* = \sigma g c^{-1} \psi_\pi$.

The PIR equilibrium is trivial and is given, as before, by $\{\hat{\pi}, \hat{x}, \hat{i}\} = \{0, 0, 0\}$. The ZIR equilibrium here is $\{\hat{\pi}, \hat{x}, \hat{i}\} = \{\frac{\mu}{\psi^*-1}, \frac{(1-\beta)\mu}{\kappa_y(\psi^*-1)}, -\mu\}$.

PIR absorbing state with active FP ($\psi^* > 1$ and $\psi^* < -\phi_\pi$). If $\psi^* > 1$ or $\psi^* < -\phi_\pi$, the economy cannot be at the ZLB in the absorbing state. That is, for an inflation level that is higher than the lower bound on the nominal interest rate, $\hat{\pi} > -\mu$, the interest rate as per the TR is unconstrained, and so the following equality – obtained by substituting (35a) into AD^{ZLB} in (35b) – implies an inflation rate that is higher that $-\mu$:

$$\hat{\pi} = \frac{\mu}{\psi^* - 1}.$$

Thus, inflation cannot be at its ZLB steady state level, and by implication the interest rate cannot be at the ZLB. Hence, no ZIR equilibrium can exist with an active FP rule as in Equation (34).

Absorbing states with passive FP ($\psi^* < 1$ and $0 < \psi^* + \phi_\pi < 1$). Here, the ZIR equilibrium is consistent with the ZLB constraint on the nominal interest rate since implied inflation is less than or equal to $-\mu$. Passive FP also implies that the slope of AD^{TR} is flatter than that of AS, hence there are two absorbing – both the PIR and ZIR equilibria. Moreover, a passive FP rule implies that in a ZIR equilibrium the inflation rate is lower than that of the ZIR equilibrium with no fiscal rule. Figure 10 illustrates the two cases.

We proceed with analysing the transitory equilibria with $\varepsilon_t = \frac{p}{\sigma}\hat{r}^T$ in each of the cases. As before, the economy remains in a transitory state for an indefinite amount of time, before transitioning to an absorbing state thereafter.

Transitory states with active FP ($\psi^* \notin (-\phi_\pi, 1)$). With probability p, the economy remains in a transitory state and jumps to the PIR absorbing state with complimentary probability $1-p$. AS and AD are given by

$$\hat{\pi}^T = \frac{\kappa_y}{1-p\beta}\hat{x}^T \; AS \tag{36a}$$

$$\hat{\pi}^T = \begin{cases} \frac{\sigma(1-p)}{c(p-\phi_\pi-\psi^*)}\hat{x}^T - \frac{p}{c(p-\phi_\pi-\psi^*)}\hat{r}^T & AD^{TR} \text{ for } \hat{\pi}^T \geq -\frac{\mu}{\phi_\pi}, \\ \frac{\sigma(1-p)}{c(p-\psi^*)}\hat{x}^T - \frac{\mu}{p-\psi^*} - \frac{p}{c(p-\psi^*)}\hat{r}^T & AD^{ZLB} \text{ for } \hat{\pi}^T \leq -\frac{\mu}{\phi_\pi}. \end{cases} \tag{36b}$$

When $\psi^* > 1$, AD^{TR} and AD^{ZLB} have negative slope. The model thus satisfies the CC conditions. When $\psi^* < -\phi_\pi$, the slopes of AD^{TR} and AD^{ZLB} are positive and AD^{ZLB} is flatter than AS. This implies a unique transitory equilibrium.

Transitory states with passive FP ($\psi^* \in (-\phi_\pi, 1)$). In the case where $\psi^* \in (-\phi_\pi, 1)$, there are two potential absorbing states. In the PIR equilibrium, we have the system as above. The slope of AD^{TR} is negative, while the slope of AD^{ZLB} can either be positive or negative. Hence, for some values of p, the slope of AD^{ZLB} can be flatter than that of AS, which implies incoherency or incompleteness in absence of support restrictions. This is the case when

$$\theta = \frac{\partial AD^{ZLB}/\partial\hat{x}}{\partial AS/\partial\hat{x}} = \frac{\sigma(1-p)(1-p\beta)}{\kappa_y c(p-\psi^*)} < 1. \tag{37}$$

If the ZIR equilibrium is the absorbing state, the system takes the form

$$\hat{\pi}^T = \frac{\kappa_y}{1-p\beta}\hat{x}^T + \frac{\beta(1-p)}{1-p\beta}\frac{\mu}{\psi^*-1}AS, \tag{38a}$$

$$\hat{\pi}^T = \begin{cases} \frac{(1-p)\sigma}{c(p-\phi_\pi-\psi^*)}\hat{x}^T - \frac{(1-p)\mu}{(\psi^*-1)(p-\psi^*-\phi_\pi)}\left[\frac{(1-\beta)\sigma}{c\kappa_y}+1\right] - \frac{p}{c(p-\phi_\pi-\psi^*)}\hat{r}^T\ AD^{TR}, \\ \frac{(1-p)\sigma}{c(p-\psi^*)}\hat{x}^T - \frac{(1-p)\mu}{(\psi^*-1)(p-\psi^*)}\left[\frac{(1-\beta)\sigma}{c\kappa_y}+1\right] - \frac{\mu}{(p-\psi^*)} - \frac{p}{c(p-\psi^*)}\hat{r}^T\ AD^{ZLB}. \end{cases} \tag{38b}$$

Since FP is passive, the slope of AD^{TR} is negative and the slope of AD^{ZLB} can be positive. In this case, the model can be incoherent or incomplete due to AD^{ZLB} being flatter than AS ($\theta < 1$).

The four cases related to active and passive FP in both absorbing states are illustrated in Figure 11.

Relationship to baseline NK model and importance of commitment. For the baseline NK model considered in 2.2.1, Equation (17) summarises the conditions under which baseline the model possesses a unique solution. It is worth reiterating that unlike in the case of a model with FP, the baseline NK model implies that the slope of AD^{ZLB} is determined only by exogenous uncertainty p and deep parameters.

Thus, one can argue that if the effects of uncertainty can be counteracted by FP, the model becomes coherent and complete. The condition on θ implies that the probability of shock persistence, p, must be low so that CC conditions are satisfied. An analogous condition for the model with permanent fiscal policy changes can be drawn from Equation (37); if $\theta > 1$, the model is coherent and complete. This condition shows that unlike in the baseline NK model, FP can alleviate the effects of exogenous uncertainty on the slope of AD^{ZLB} and, thus, ensure satisfaction of CC conditions.

The importance of persistence implied by (34) cannot be overstated and is a key point

of this paper. To highlight it, consider the case where the fiscal targeting rule is given in deviations and not in growth rates, i.e. $\widehat{GR}_t = \psi_\pi \hat{\pi}_t$. This will imply the following AD^{ZLB}/AS slope ratio that is required to be greater than unity to satisfy CC:

$$\theta'' = \frac{\sigma(1-p)(1-p\beta)}{p+\psi^*(1-p)} > 1,$$

which does not hold if p is sufficiently large for any bounded value of ψ^*.

The key intuition for permanent policy changes can be drawn from the fact that the FP instrument is present in the aggregate demand relationship in Equation (30) in expectation. Thus, any rule that targets contemporaneous deviations would imply additional terms in expectation that add uncertainty to the system. This highlights the importance of commitment to future changes in policy that depend on contemporaneous deviations of endogenous variables as, for example, in (34). The assumption that a fiscal authority needs to apply a targeting rule to growth rates of the instrument and not to its deviations is rather restrictive. We relax this assumption in Section 4.1 allowing for an inertial rule.

3.2 Equivalence of Simple Fiscal Policy Regimes

So far, we have considered a standard NK model augmented with a simple FP setup where government spending is financed by lump-sum and output taxes. The key difference between this model, described by the system (19), and a standard NK model as in Equations (1a)-(1c) is the presence of a fiscal instrument in the DISE and not in the NKPC. As shown above, if FP uses this instrument to react to exogenous disturbances aggressively enough, the model satisfies the CC conditions. Existence of such an instrument in the DISE is, however, not exclusive to the fiscal setup we have discussed.

For example, consider the case where the fiscal authority levies consumption and wage taxes, τ^c and τ^w, respectively, and only redistributes the taxes as lump-sum transfers, τ_t. Additionally, assume that there are no production subsidies, $\tau_t^s = 0, \forall t$. Then, if the following condition holds, FP instruments do not show up in the NKPC:

$$\Psi^w \hat{\tau}_t^w = \Psi^c \hat{\tau}_t^c. \tag{39}$$

The natural interest rate in the DISE can then be written as:

$$\hat{r}_t^n = -\mathbb{E}_t \Delta \hat{\tau}_{t+1}^c - \frac{\sigma}{c} \varepsilon_t \tag{40}$$

where $\Delta \hat{\tau}_{t+1}^c$ is the consumption tax growth rate.

It is straightforward to infer that FP can replace active monetary policy when the former is constrained (see, for example, the "knife-edge" case) thus rendering the model linear and guaranteeing that the CC conditions are satisfied. This is in line with the results in Correia et al. (2013). Under this formulation, the strength of the fiscal instrument may be higher or lower than the DISE in (19), depending on the values of the model's structural parameters. For example, under our calibration in Table 1, one can observe that the coefficient on the fiscal instrument in the latter is g, while on the former it is Ψ^c/σ. While qualitatively the role of the instruments in both cases is identical, the degree of reaction of $\Delta \hat{\tau}_{t+1}^c$ is required to be greater than that of $\Delta \widehat{GR}_{t+1}$ since the coefficient on the former is smaller.

Using this alternative setup, we show the CC regions in Figure 12 under: (i) simple inflation targeting in Figure 12a and (ii) contemporaneous inflation and output targeting in Figure 12b. The relevant coefficients of the canonical form are given in Appendix B.4.

As before, if the degree of reaction of differentials of consumption tax to inflation is large enough, the model is coherent and complete. The intuition for this case is simple and mirrors that where the only distortionary tax was the output tax.

4 Coherency and Completeness with an Endogenous State

Thus far, we have only considered the case of a simple NK-FP model that did not feature endogenous state variables. This was to keep the computation and verification of the CC conditions analytically tractable. Namely, we assume $\boldsymbol{H}_{s_t} = \boldsymbol{O}$ and $\boldsymbol{h} = \mathbf{0}$ in (2). With these assumptions, an MSV solution can be represented as $\boldsymbol{Y}_t = f(\boldsymbol{X}_t)$ by the time-invariant matrix $\mathbf{Y}$; and variables along a MSV solution satisfy:

$$\mathbb{E}_t\left[\boldsymbol{Y}_{t+1}|\boldsymbol{Y}_t = \mathbf{Y}\boldsymbol{e}_i\right] = \mathbb{E}_t\left[\boldsymbol{Y}_{t+1}|\boldsymbol{X}_t = \mathbf{X}\boldsymbol{e}_i\right] = \mathbf{Y}\boldsymbol{K}^\top\boldsymbol{e}_i.$$

However, this assumption is restrictive and makes assessment of CC conditions in conventional fiscal policy-equipped DSGE models limited, particularly in the literature exploring the monetary and fiscal policy mix and ZLB. In this section, we loosen this assumption and consider a canonical NK-FP model with an endogenous state variable, namely taxes.

Thus, we no longer assume that $\boldsymbol{H}_{s_t} = \boldsymbol{O}$ and $\boldsymbol{h} = \mathbf{0}$ in (2). We maintain the assumption that $\boldsymbol{X}_t$ follows a k-state stationary Markov process. This implies that, as before, the i-th column of $\mathbf{X}$ gives the value of $\boldsymbol{X}_t$ for a given state i. However, as stipulated by AM, with endogenous states the support of $\boldsymbol{Y}_t$ will vary endogenously over time along the MSV solution given by $\boldsymbol{Y}_t = f(\boldsymbol{Y}_{t-1}, \boldsymbol{X}_t)$. This implies that the solution can no longer be characterised by a time invariant matrix $\mathbf{Y}$. In other words, despite

the variables $\boldsymbol{X}_t$ being time invariant (by definition as they are purely forward looking), the support of $\boldsymbol{Y}_t$ must now be a function of $\boldsymbol{Y}_{t-1}$, too. With endogenous states, along a MSV solution we have:

$$\mathbb{E}_t\left[\boldsymbol{Y}_{t+1}|\boldsymbol{Y}_t = \mathbf{Y}_t\boldsymbol{e}_i, \boldsymbol{X}_t = \mathbf{X}\boldsymbol{e}_i\right] = \mathbf{Y}_{t+1}^i\boldsymbol{K}^\top\boldsymbol{e}_i,$$

where $\mathbf{Y}_{t+1}^i$ gives the support of $\boldsymbol{Y}_{t+1}$ when $\boldsymbol{Y}_t$ is in the i-th state. However, this presents a non-trivial computational challenge: the support of $\boldsymbol{Y}_t$ is exponentially rising for a given initial condition $\boldsymbol{Y}_0$. Therefore, a MSV solution cannot be characterised by a finite-dimensional system of piecewise linear equations. This requires a different method of analysis to that of Section 3.

In the fashion of AM, we solve the model recursively from some terminal state $t = T$. For simplicity, assume that the endogenous state variable is a scalar, $\boldsymbol{H}_{s_t}\boldsymbol{Y}_{t-1} = \boldsymbol{h}_{s_t}y_{t-1}$, where $\boldsymbol{h}_{s_t}$ is $n \times 1$ and $y_t = \boldsymbol{g}^\top\boldsymbol{Y}_t$ is a linear combination of $\boldsymbol{Y}_t$ and where $\boldsymbol{g} = \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}^\top$.

For a date T whereby $t \geq T$, the MSV solution $f(y_{t-1}, \boldsymbol{X}_t)$ can be written as

$$\mathbf{Y}_t = \boldsymbol{G}y_{t-1} + \boldsymbol{Z},$$

where $\boldsymbol{G}$ and $\boldsymbol{Z}$ are $n \times k$ matrices. $\boldsymbol{Z}$ captures the portion of $\mathbf{Y}_t$ that depends on exogenous variables $\boldsymbol{X}_t$. In the case of no exogenous variables, we have that $\boldsymbol{G} = \boldsymbol{O}$ and so $\mathbf{Y}_t = \boldsymbol{Z}$, yielding the standard case with a time invariant $\mathbf{Y}$, and the analysis of Section 3 applies. The columns of $\boldsymbol{G}$ return the coefficients on y_{t-1} in the MSV solution, mapping it to different states of $\boldsymbol{X}_t$. Assume, as before, that $k = 2$, whereby the "bad state" corresponds to $i = 1$, and the"good state" is given by $i = 2$. In other words, $i = 1$

is ZIR state and $i = 2$ is a PIR state. Then the endogenous dynamics in the bad state can be different from the good state.

With no endogenous dynamics, where $\mathbf{Y} = \boldsymbol{Z}$, we can put the model in canonical form as in (3). With endogenous dynamics, the equivalent expression is given by

$$\begin{aligned} \mathbf{0} = &\left(\boldsymbol{A}_{s_{t,i}}\boldsymbol{G}\boldsymbol{e}_i + \boldsymbol{h}_{s_{t,i}} + \boldsymbol{B}_{s_{t,i}}\boldsymbol{G}\boldsymbol{K}^\top\boldsymbol{e}_i\boldsymbol{g}^\top\boldsymbol{G}\boldsymbol{e}_i\right) y_{t-1} \\ &+ \left(\boldsymbol{A}_{s_{t,i}}\boldsymbol{Z} + \boldsymbol{B}_{s_{t,i}}\boldsymbol{G}\boldsymbol{K}^\top\boldsymbol{e}_i\boldsymbol{g}^\top\boldsymbol{Z} + \boldsymbol{B}_{s_{t,i}}\boldsymbol{Z}\boldsymbol{K}^\top + \boldsymbol{C}_{s_{t,i}}\mathbf{X} + \boldsymbol{D}_{s_{t,i}}\mathbf{X}\boldsymbol{K}^\top\right)\boldsymbol{e}_i, \end{aligned} \tag{41}$$

for all $i = 1, ..., k$. For a given regime J corresponding to the k states and their transitions, $\{(PIR, PIR), (ZIR, PIR), (PIR, PIR), (ZIR, ZIR)\}$ (see (8)), a slackness condition for the constraint $s_{t,i}$ is determined which gives a system of $2nk$ polynomial equations in the $2nk$ unknowns $\boldsymbol{G}$ and $\boldsymbol{Z}$ by equating the coefficients on y_{t-1} and the constant terms to zero, respectively. As these conditions are polynomial and not piecewise linear in $\boldsymbol{G}$ and $\boldsymbol{Z}$, the algorithm and theorem of Gourieroux, Laffont, and Monfort (1980) is no longer suitable to check coherency. Instead, we build on the algorithm and "brute force" numerical solution method of AM,[16] which essentially goes through all possible 2^k J regime configurations to check if there are any feasible solutions that satisfy the inequality constraints.

The model can be solved backwards starting from some terminal date T. We know that at T, the solution to the model takes the following form

$$\boldsymbol{Y}_T = \boldsymbol{G}_{J_0} y_{T-1} + \boldsymbol{Z}_{J_0},$$

where $J_0 \in J$ defines the configuration of regimes in T. $\boldsymbol{G}_{J_0}$ and $\boldsymbol{Z}_{J_0}$ can be solved for

16. See Appendix A.5.2 of their paper.

from (41):

$$\mathbf{0} = \boldsymbol{A}_{s_{t,i}}\boldsymbol{G}\boldsymbol{e}_i + \boldsymbol{h}_{s_{t,i}} + \boldsymbol{B}_{s_{t,i}}\boldsymbol{G}\boldsymbol{K}^\top\boldsymbol{e}_i\boldsymbol{g}^\top\boldsymbol{G}\boldsymbol{e}_i, \tag{42}$$

$$\mathbf{0} = \left(\boldsymbol{A}_{s_{t,1}}\boldsymbol{Z} + \boldsymbol{B}_{s_{t,i}}\boldsymbol{G}\boldsymbol{K}^\top\boldsymbol{e}_i\boldsymbol{g}^\top\boldsymbol{Z} + \boldsymbol{B}_{s_{t,i}}\boldsymbol{Z}\boldsymbol{K}^\top + \boldsymbol{C}_{s_{t,i}}\mathbf{X} + \boldsymbol{D}_{s_{t,i}}\mathbf{X}\boldsymbol{K}^\top\right)\boldsymbol{e}_i, \tag{43}$$

$\forall i = 1, ..., k$.

$\boldsymbol{Y}_T$ is a function of $\boldsymbol{G}_{J_0}$ and $\boldsymbol{Z}_{J_0}$. Thus, $\boldsymbol{Y}_T$ is known and we can solve for $\boldsymbol{Y}_{T-1}$ from

$$\begin{aligned}\mathbf{0} = &\left(\boldsymbol{A}_{s_{T-1},i} + \boldsymbol{B}_{s_{T-1},i}\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_i\boldsymbol{g}^\top\boldsymbol{Y}_{T-1}\boldsymbol{e}_i\right)\\ &+ \left(\boldsymbol{B}_{s_{T-1},i}\boldsymbol{Z}_{J_0}\boldsymbol{K}^\top + \boldsymbol{C}_{s_{T-1},i}\mathbf{X} + \boldsymbol{D}_{s_{T-1},i}\mathbf{X}\boldsymbol{K}^\top\right)\boldsymbol{e}_i + \boldsymbol{h}_{s_{T-1},i}y_{T-2}.\end{aligned}$$

For every t the determinants relevant for CC conditions are given by

$$|\mathcal{A}_{J_0J_1}| = \prod_i^k \det\left(\boldsymbol{A}_{s_{T-1},i} + \boldsymbol{B}_{s_{T-1},i}\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_i\boldsymbol{g}^\top\right).$$

If $k = 2$, the determinants can be rewritten as

$$\begin{aligned}|\mathcal{A}_{J_0\{1,2\}}| &= \det\left(\boldsymbol{A}_1 + \boldsymbol{B}_1\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_1\boldsymbol{g}^\top\right)\det\left(\boldsymbol{A}_1 + \boldsymbol{B}_1\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_2\boldsymbol{g}^\top\right),\\ |\mathcal{A}_{J_0\{2\}}| &= \det\left(\boldsymbol{A}_0 + \boldsymbol{B}_0\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_1\boldsymbol{g}^\top\right)\det\left(\boldsymbol{A}_1 + \boldsymbol{B}_1\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_2\boldsymbol{g}^\top\right),\\ |\mathcal{A}_{J_0\{1\}}| &= \det\left(\boldsymbol{A}_0 + \boldsymbol{B}_0\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_1\boldsymbol{g}^\top\right)\det\left(\boldsymbol{A}_1 + \boldsymbol{B}_1\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_2\boldsymbol{g}^\top\right),\\ |\mathcal{A}_{J_0\{\emptyset\}}| &= \det\left(\boldsymbol{A}_0 + \boldsymbol{B}_0\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_1\boldsymbol{g}^\top\right)\det\left(\boldsymbol{A}_0 + \boldsymbol{B}_0\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_2\boldsymbol{g}^\top\right).\end{aligned}$$

If the model satisfies CC, the solution is given by

$$\begin{aligned}\boldsymbol{Y}_{T-1}\boldsymbol{e}_i = &-\left(\boldsymbol{A}_{s_{T-1},i} + \boldsymbol{B}_{s_{T-1},i}\boldsymbol{G}_{J_0}\boldsymbol{K}^\top\boldsymbol{e}_i\boldsymbol{g}^\top\right)^{-1}\\ &\left[\left(\boldsymbol{B}_{s_{T-1},i}\boldsymbol{Z}_{J_0}\boldsymbol{K}^\top + \boldsymbol{C}_{s_{T-1},i}\mathbf{X} + \boldsymbol{D}_{s_{T-1},i}\mathbf{X}\boldsymbol{K}^\top\right)\boldsymbol{e}_i + \boldsymbol{h}_{s_{T-1},i}y_{T-2}\right],\end{aligned}$$

$\forall i = 1, ..., k$.

Iterating the solution backwards implies that all the determinants $\mathcal{A}_{J_0,...,J_{T-t}}$ must have the same sign. The recursive solution will be given by

$$\boldsymbol{Y}_t = \boldsymbol{G}_{J_0,...,J_{T-t}} y_{t-1} + \boldsymbol{Z}_{J_0,...,J_{T-t}},$$

where $\boldsymbol{G}_{J_0,...,J_{T-t}}$ and $\boldsymbol{Z}_{J_0...,J_{T-t}}$ can be computed recursively using

$$\begin{aligned} \boldsymbol{Z}_{J_0...,J_{T-t},i} = &- \left(\boldsymbol{A}_{s_{t,i}} + \boldsymbol{B}_{s_{t,i}} \boldsymbol{G}_{J_0,...,J_{T-t-1}} \boldsymbol{K}^\top \boldsymbol{e}_i \boldsymbol{g}^\top\right)^{-1} \\ &\left(\boldsymbol{B}_{s_{t,i}} \boldsymbol{Z}_{J_0,...,J_{T-t-1}} \boldsymbol{K}^\top + \boldsymbol{C}_{s_{t,i}} \mathbf{X} + \boldsymbol{D}_{s_{t,i}} \mathbf{X} \boldsymbol{K}^\top\right) \boldsymbol{e}_i, \end{aligned} \tag{44}$$

$$\boldsymbol{G}_{J_0,...,J_{T-t},i} = - \left(\boldsymbol{A}_{s_{t,i}} + \boldsymbol{B}_{s_{t,i}} \boldsymbol{G}_{J_0,...,J_{T-t-1}} \boldsymbol{K}^\top \boldsymbol{e}_i \boldsymbol{g}^\top\right)^{-1} \boldsymbol{h}_{s_{t,i}}. \tag{45}$$

The recursive solution from terminal T solves the model backwards to $t = 1$ and implies up to $2^{(T-1)k}$ solution paths. Given some initial condition, y_0, and conditional on satisfaction of CC conditions, the recursive solution is unique. If the CC conditions are not satisfied, there can be either no or multiple solutions.

We thus apply the following algorithm to check CC conditions assuming that the shocks are two-state Markovian. First, calculate $\boldsymbol{G}_{J_0}$ and $\boldsymbol{Z}_{J_0}$ from (42) and (43) for four possible regime configurations in J_0. For each of the four regime configurations in J_1 compute $|\mathcal{A}_{J_0 J_1}|$. If for some regime configuration J_0, $|\mathcal{A}_{J_0 J_1}|$ have the same sign, a unique solution is possible; or else we conclude that no unique solution exists. Second, for all configurations of J_0, where $|\mathcal{A}_{J_0 J_1}|$ have the same sign, compute $\boldsymbol{G}_{J_0 J_1}$ and $\boldsymbol{Z}_{J_0 J_1}$ using (44) and (45). Third, continue solving backwards for each J_{T-t} until $t = 1$. If $t = 1$ can be reached with: i) the condition on the signs of determinants being satisfied along the solution path; and ii) the model solution being consistent with the implied $s_{t,i}, \forall t$,

then we can conclude that the model is coherent and complete.

4.1 New Keynesian Model with Fiscal Persistence

We have shown in Section 3.1 how a permanent change in fiscal policy can maintain coherency and completeness. In this section, we generalise over the fiscal rule used previously and allow for inertia instead of a permanent policy change.

Consider the baseline NK-FP model described by (19). But assume that there are no distortionary taxes ($\hat{\tau}_t^s = \hat{\tau}_t^c = \hat{\tau}_t^w = 0, \forall t$), and abstract from TFP shocks ($\hat{a}_t = 0, \forall t$) so that the natural interest rate is given by (21). However, we close the model by augmenting the rule (22) to account for inertia in $\widehat{GR}_t$. Specifically, we replace (22) with:

$$\widehat{GR}_t = \rho_\tau \widehat{GR}_{t-1} + \psi_\pi \hat{\pi}_t + \psi_y \hat{x}_t, \tag{46}$$

where $\rho_\tau \in (0, 1)$ is a persistence parameter. Thus, the system of equations considered is:

$$\hat{x}_t = \mathbb{E}_t \hat{x}_{t+1} - \frac{c}{\sigma}\left(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1} - \hat{r}_t^n\right), \tag{47a}$$

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_y \hat{x}_t - \kappa_g \widehat{GR}_t, \tag{47b}$$

$$\hat{i}_t = \max\{-\mu; \phi_\pi \hat{\pi}_t + \phi_y \hat{x}_t\}, \tag{47c}$$

$$\widehat{GR}_t = \rho_\tau \widehat{GR}_{t-1} + \psi_\pi \hat{\pi}_t + \psi_y \hat{x}_t, \tag{47d}$$

with

$$\hat{r}_t^n = -\frac{\sigma}{c}(g \mathbb{E}_t \Delta \widehat{GR}_{t+1} + \varepsilon_t).$$

The system in (47) can be evaluated about two absorbing states, either PIR or ZIR.

About the PIR steady state, it must be the case that $\{\hat{x}, \hat{\pi}, \hat{i}, \widehat{GR}\} = \{0, 0, 0, 0\}$. The existence of the PIR steady state is trivial to reconcile. When policy is effective, the inflation and output gaps are closed and the system of equations gives the solution in the PIR absorbing state. However, about the ZIR absorbing state we have that $\hat{i} = -\mu$, thus giving a solution of the form

$$\{\hat{x}, \hat{\pi}, \hat{i}, \widehat{GR}\} = \left\{ -\frac{(1-\beta)(1-\rho_\tau) + \kappa_g \psi_\pi}{\kappa_y(1-\rho_\tau) - \kappa_g \psi_y} \mu, -\mu, -\mu, \frac{\psi_y \hat{x} + \psi_\pi \hat{\pi}}{1-\rho_\tau} \right\}. \tag{48}$$

One can observe that under certain fiscal policy rules, the above ZIR equilibrium is not consistent with the constraint on the TR and, thus, the ZIR equilibrium is ruled out. Specifically, we require that $\hat{x}$ be sufficiently large such that the ZLB constraint on $\hat{i}_t$ is not binding:

$$-\mu < -\phi_\pi \mu + \phi_y \hat{x} \implies -\frac{\mu(1-\phi_\pi)}{\phi_y} < \hat{x}, \quad \phi_y \neq 0.$$

This implies

$$\frac{1-\phi_\pi}{\phi_y} > \frac{(1-\beta)(1-\rho_\tau) + \kappa_g \psi_\pi}{\kappa_y(1-\rho_\tau) - \kappa_g \psi_y},$$

with the LHS being negative under conventional restrictions on TR coefficients. As ρ_τ tends to unity, we get

$$\lim_{\rho_\tau \to 1} \frac{(1-\beta)(1-\rho_\tau) + \kappa_g \psi_\pi}{\kappa_y(1-\rho_\tau) - \kappa_g \psi_y} = -\frac{\psi_\pi}{\psi_y} < \frac{1-\phi_\pi}{\phi_y},$$

which holds under countercyclical fiscal policy, $\psi_y < 0$, $\psi_\pi < 0$ with ψ_π sufficiently large in absolute value. Thus, under countercyclical fiscal policy, if ρ_τ is sufficiently large there cannot exist a ZIR absorbing state. This analysis highlights two points made earlier in Section 3. First, if the monetary authority targets the output gap, a ZIR equilibrium can

be ruled out. Second, if the fiscal rule is sufficiently persistent, the ZIR absorbing state is ruled out under countercyclical fiscal policy.

As the model contains an endogenous state, it cannot be represented as a finite-dimensional piecewise linear function and the GLM result does not apply. We are limited in using analytical expressions to exhibit intuition; the CC conditions cannot be verified analytically. Thus, we have the following proposition that we verify numerically using an algorithm that is based on the work by AM:[17]

Proposition 5. *A New Keynesian model subject to an occasionally binding ZLB constraint on interest rates and with fiscal policy as described in* (47) *satisfies the coherency and completeness conditions if fiscal policy is sufficiently persistent and countercyclical.*

The system of equations in (47) can be cast in the canonical form (41) with relevant coefficients given in Appendix B.5.

We plot regions where CC and BK local determinacy conditions are satisfied in Figure 13 for different values of ρ_τ. As fiscal policy becomes more persistent, the CC region becomes larger. Moreover, the figure shows that the parameter space where CC conditions are satisfied largely corresponds to regions where the model satisfies BK conditions (blue circles).

To satisfy CC, the fiscal policy is required to be sufficiently persistent and aggressive. First, the intuition for persistence can be drawn from the model considered in Subsection 3.1.1. There, we considered the special case where $\rho_\tau = 1$, which implied that the fiscal authority commits to a permanent policy change in reaction to deviations of inflation and the output gap. Committing to the policy change in this case can be seen as a form of expectations management as the fiscal authority promises to change spending

17. AM developed an algorithm to verify the CC conditions for baseline New Keynesian model subject to the ZLB on interest rates, active TR, and whereby the TR exhibited persistence.

permanently in response to low inflation. This promise can be seen as a factor that reduces fundamental uncertainty in the system. Second, the policy is required to be sufficiently aggressive to guarantee that its effect on the system is sufficiently large to eliminate multiplicity of equilibria. Much like the Taylor principle requires the monetary authority to react by more than one-to-one to inflation, we require that fiscal policy is sufficiently aggressive to guarantee a unique solution.

Other approaches in the literature (price level targeting (PLT) in Holden (2022) and unconventional monetary policy (UMP) in AM and Ikeda et al. (2021)) rely on a similar mechanism to guarantee uniqueness. As argued in Holden (2022), PLT rules can restore uniqueness in the presence of an occasionally binding ZLB constraint as such a policy implies a promise about future inflation given inflation today. If MP is committed to a given price level path, the monetary authority promises that a period of low inflation today will be followed by a period of high inflation in the future. Thus, agents expecting high prices in the future increase their consumption in periods of low inflation and, by implication, the system has a unique solution around the PIR absorbing state. The commitment to higher inflation in the future delivers sufficient information about the expected dynamics of the system that alleviates uncertainty that would otherwise engender multiplicity and, by implication, pins down the unique solution much like persistent fiscal policy.

As shown in AM, the baseline NK model with UMP as in Chen, Cúrdia, and Ferrero (2012) satisfies the CC conditions if UMP is effective enough. This result is consistent with the logic presented above. When the ZLB is binding, the effect of UMP on model dynamics needs to be sufficiently strong to pin down a unique solution. In this case, UMP is used to alleviate the effects of exogenous uncertainty and ensure that the solution is unique around the PIR absorbing state. This is in line with the restrictions we establish

for FP such that it guarantees a unique solution.

5 Conclusion

This paper explores whether fiscal policy can restore coherency and completeness in a baseline NK model subject to an occasionally binding constraint on the interest rate that is generally incoherent in absence of FP. Our findings suggest that simple Ricardian FP can restore coherency and completeness thus guaranteeing a unique solution that is also locally determinate. Fiscal policy that guarantees a unique global and local solution needs to be sufficiently persistent and aggressive.

First, we analytically verify that if the fiscal authority is able to credibly commit to a sufficiently strong permanent policy change in response to an exogenous disturbance, coherency and completeness of the model is restored. This conclusion is rationalised by the fact that fiscal policy is not constrained by the ZLB and provides an active policy response when monetary policy is constrained. Moreover, by committing to a permanent policy change, the fiscal authority is able to alleviate the fundamental uncertainty that engenders multiplicity of equilibria in the baseline NK model.

Second, we find that the fiscal response need not imply a permanent policy change but rather it has to be sufficiently persistent to guarantee existence and uniqueness. The persistence property of the policy rule coupled with it being sufficiently countercyclical are needed to eliminate uncertainty-driven multiplicity of equilibria and pin down a unique solution. By showing this, we address the main concerns raised by Ascari and Mavroeidis (2022) about NK models that feature occasionally binding constraints.

Despite this paper being a theoretical contribution to the literature, it offers a rationalisation for recent world events that saw fiscal authorities engage in large-scale stimulus

programmes in response to the sharp contraction caused by the COVID-19 pandemic. Our result adds to the array of methods that have been found to guarantee model solution uniqueness in the presence of the ZLB: price level targeting and unconventional monetary policy. The implications of using both UMP and persistent countercyclical fiscal policy for coherency and completeness presents a promising avenue for future research.

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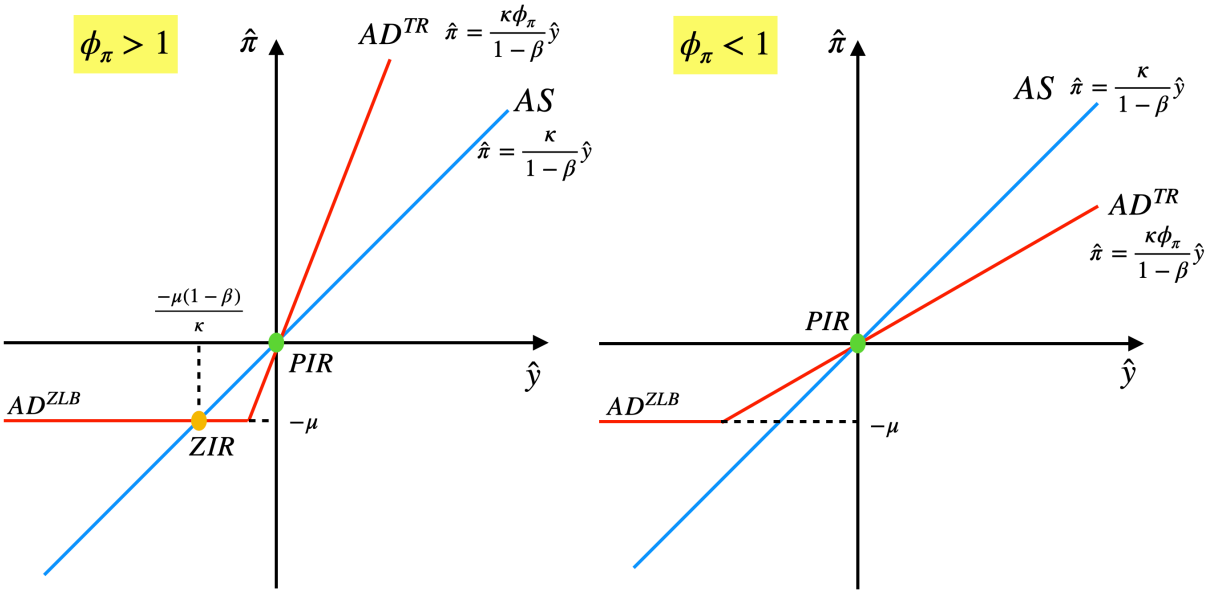
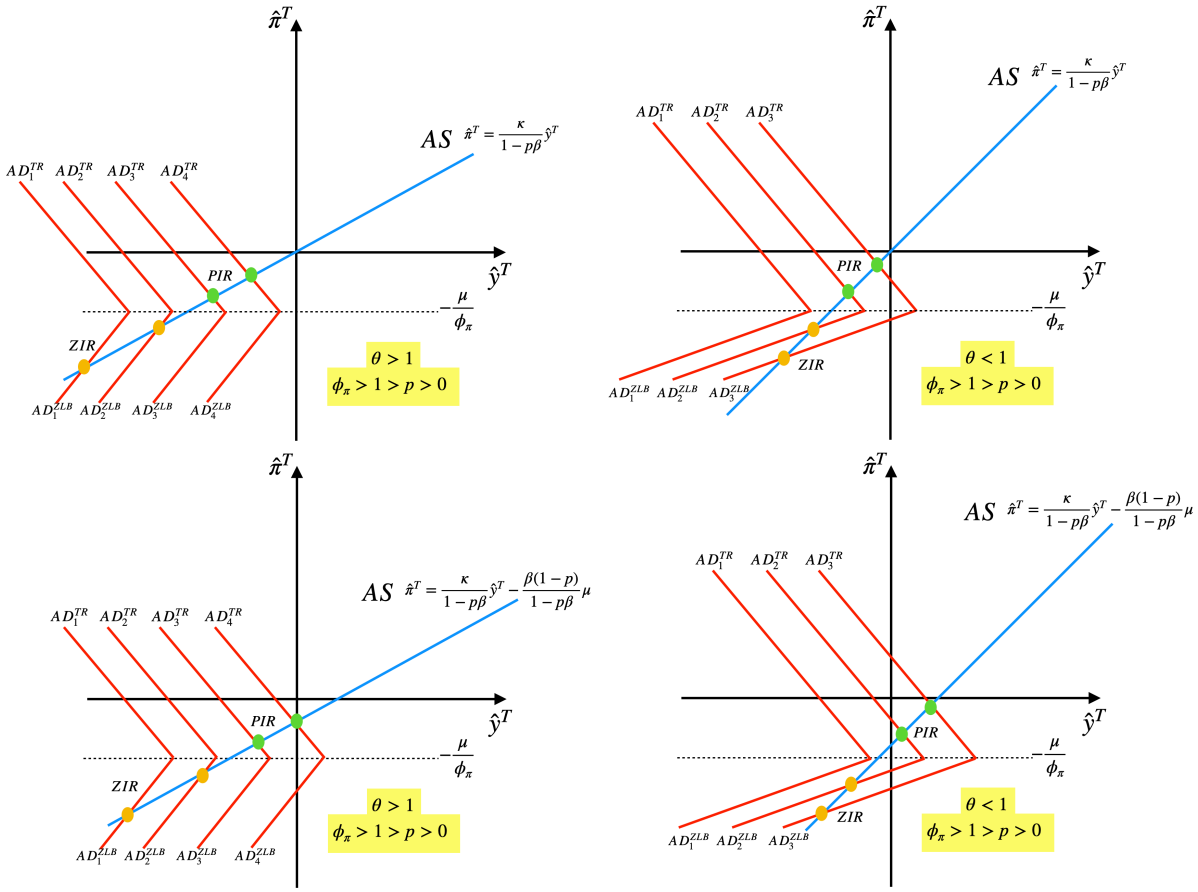
Figure 1: Absorbing State of the New Keynesian Model ($\varepsilon_t = 0$)



Note: Diagram on the left depicts equilibria when the Taylor principle is adhered to. Diagram on the right shows the PIR absorbing state for an interest rate rule that does not satisfy the Taylor principle.

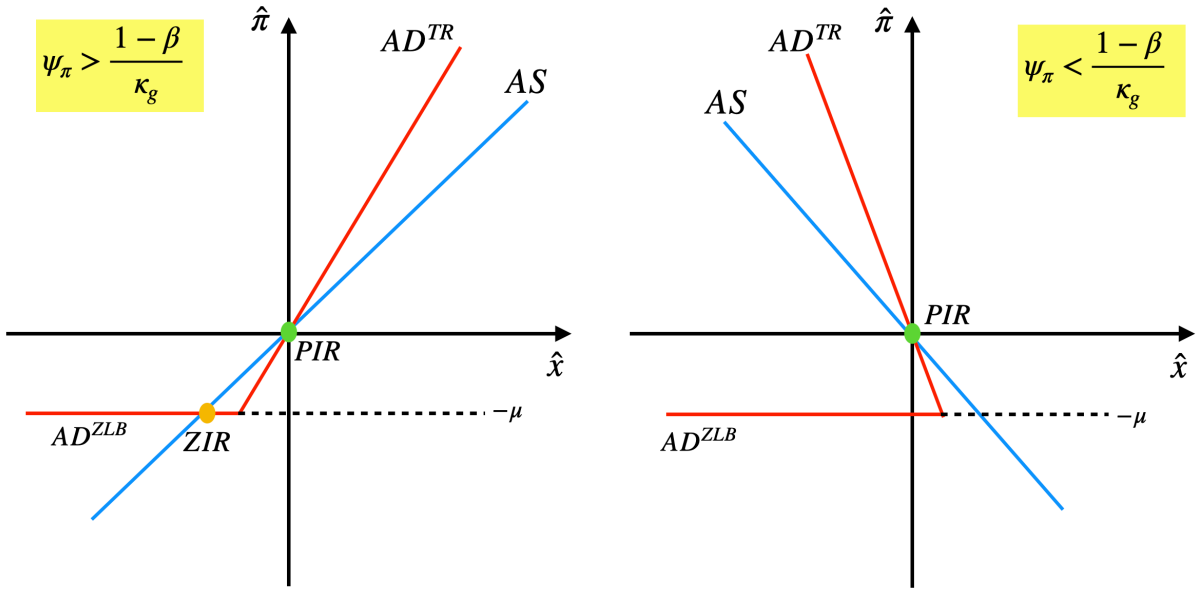
Figure 2: Transitory State of the New Keynesian Model ($\phi_\pi > 1$)



(a) PIR (top) and ZIR (bottom) Absorbing States ($\theta > 1$)

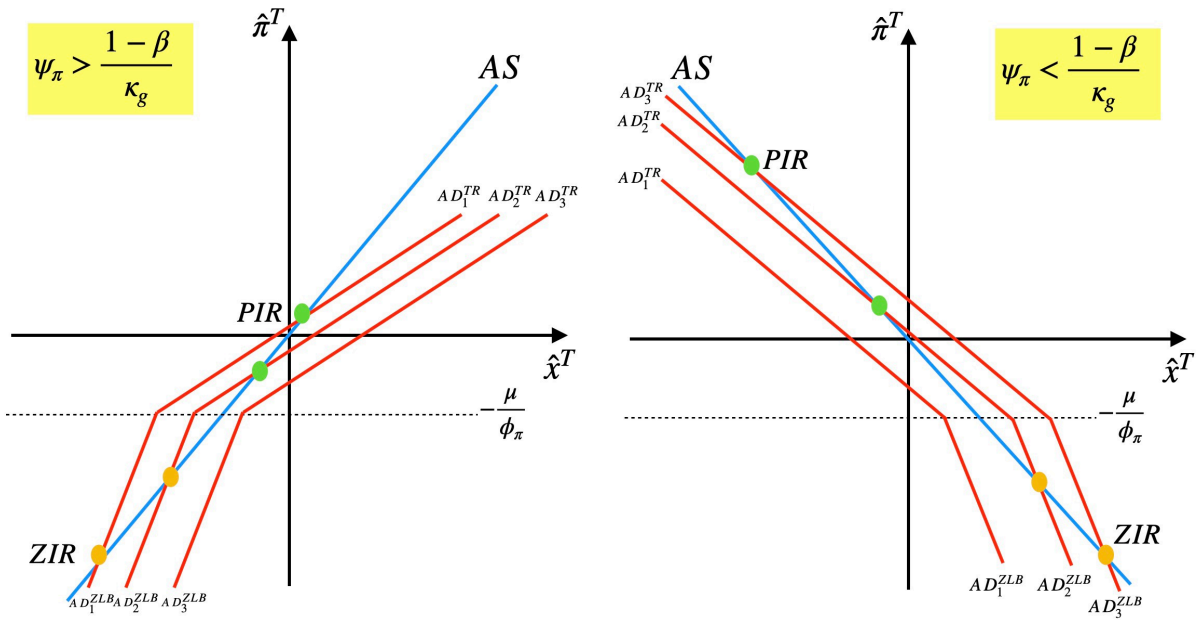
(b) PIR (top) and ZIR (bottom) Absorbing States ($\theta < 1$)

Figure 3: Strict Inflation Targeting MP: Absorbing State



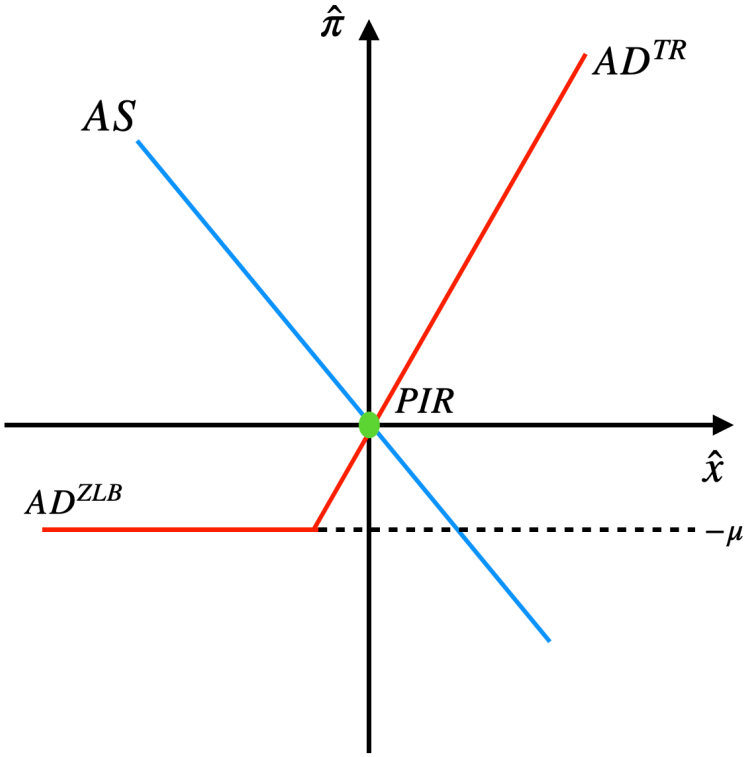
Note: The absorbing state is described as the permanent state of the economy when $\varepsilon_t = 0$. Monetary policy is conducted by adjusting the nominal interest rate to only close the inflation gap, implying that $\phi_y = 0$.

Figure 4: Strict Inflation Targeting MP: Transitory State with PIR Absorbing



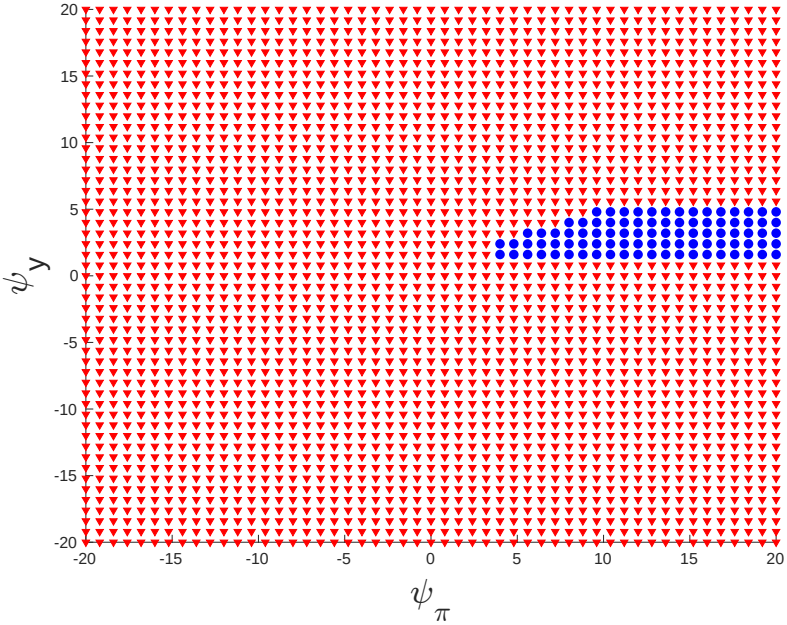
Note: In each period the economy is subject to a shock with probability p. With complementary probability $1-p$, the economy transitions to the PIR absorbing state. Monetary policy is conducted by adjusting the nominal interest rate to only close the inflation gap, implying that $\phi_y = 0$.

Figure 5: Inflation and Outgap Gap Targeting MP



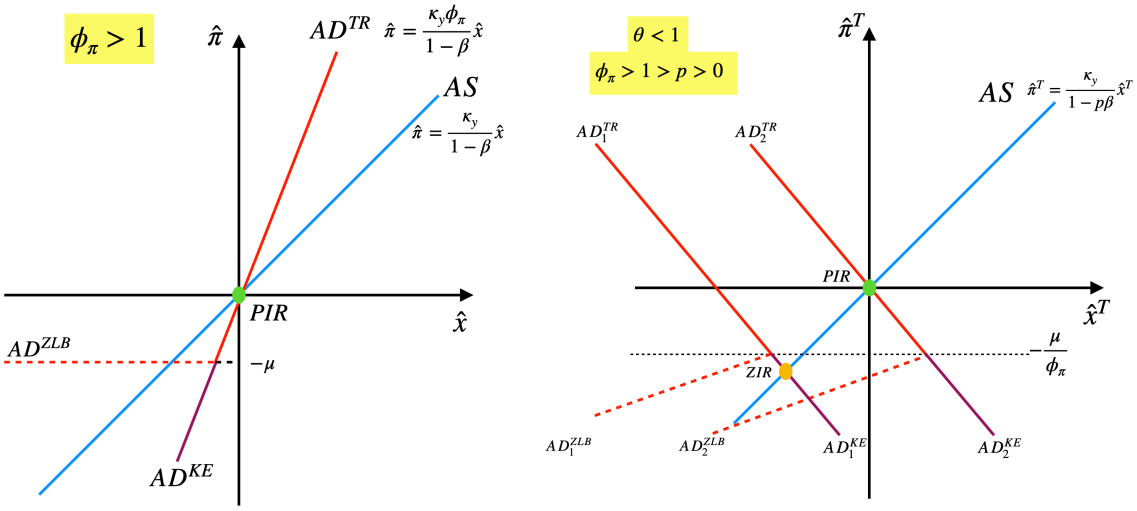
Note: Figures hows when monetary policy is conducted by adjusting the nominal interest rate to both the inflation and output gap, and illustrates the case where the economy is not subject to shocks ($\varepsilon_t = 0$).

Figure 6: Coherency and Completeness Region for Inflation and Output Gap Targeting MP



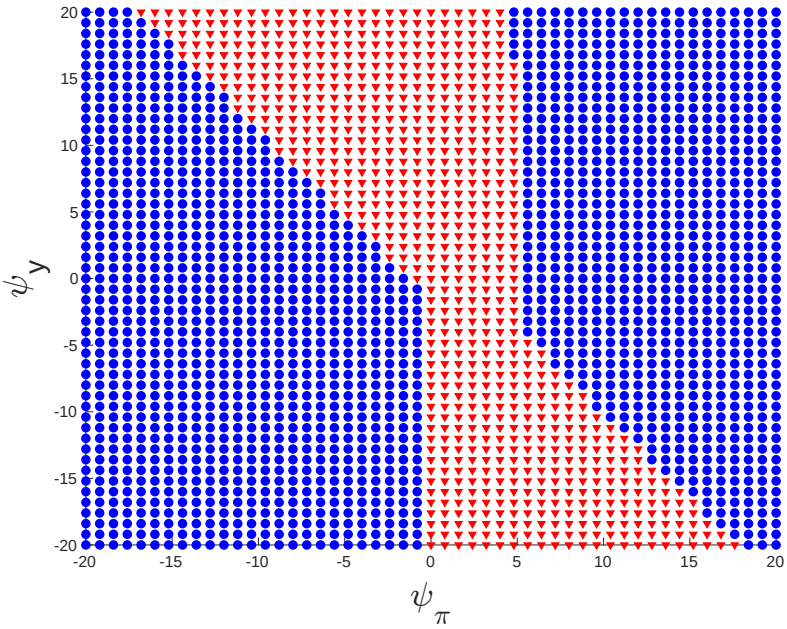
Note: Blue circles denote regions where coherency and completeness conditions are satisfied. Red triangles denotes region where the model is either incoherent or incomplete.

Figure 7: Coherency and Completeness with Knife-Edge Spending Rule



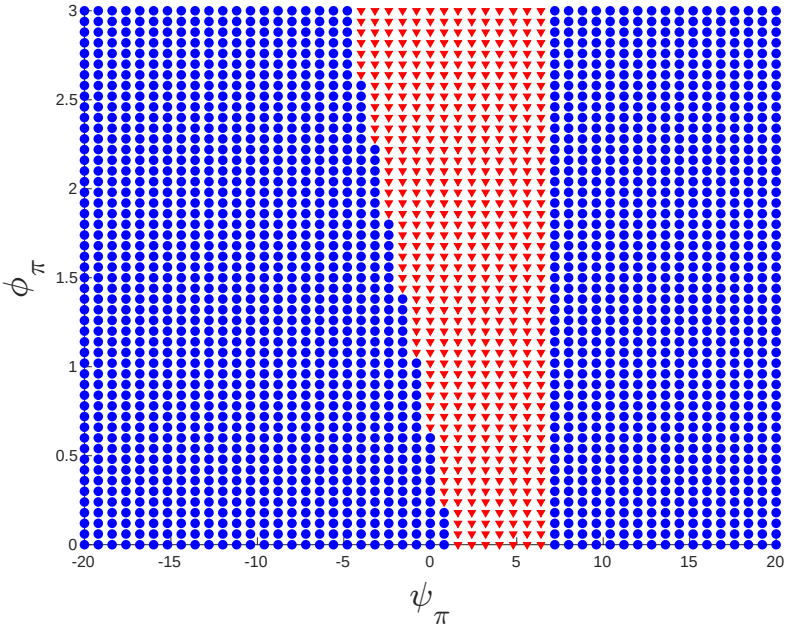
(a) Absorbing State (b) Transitory State with PIR Absorbing

Figure 8: Coherency and Completeness Region for Inflation and Output Gap Fiscal Rule



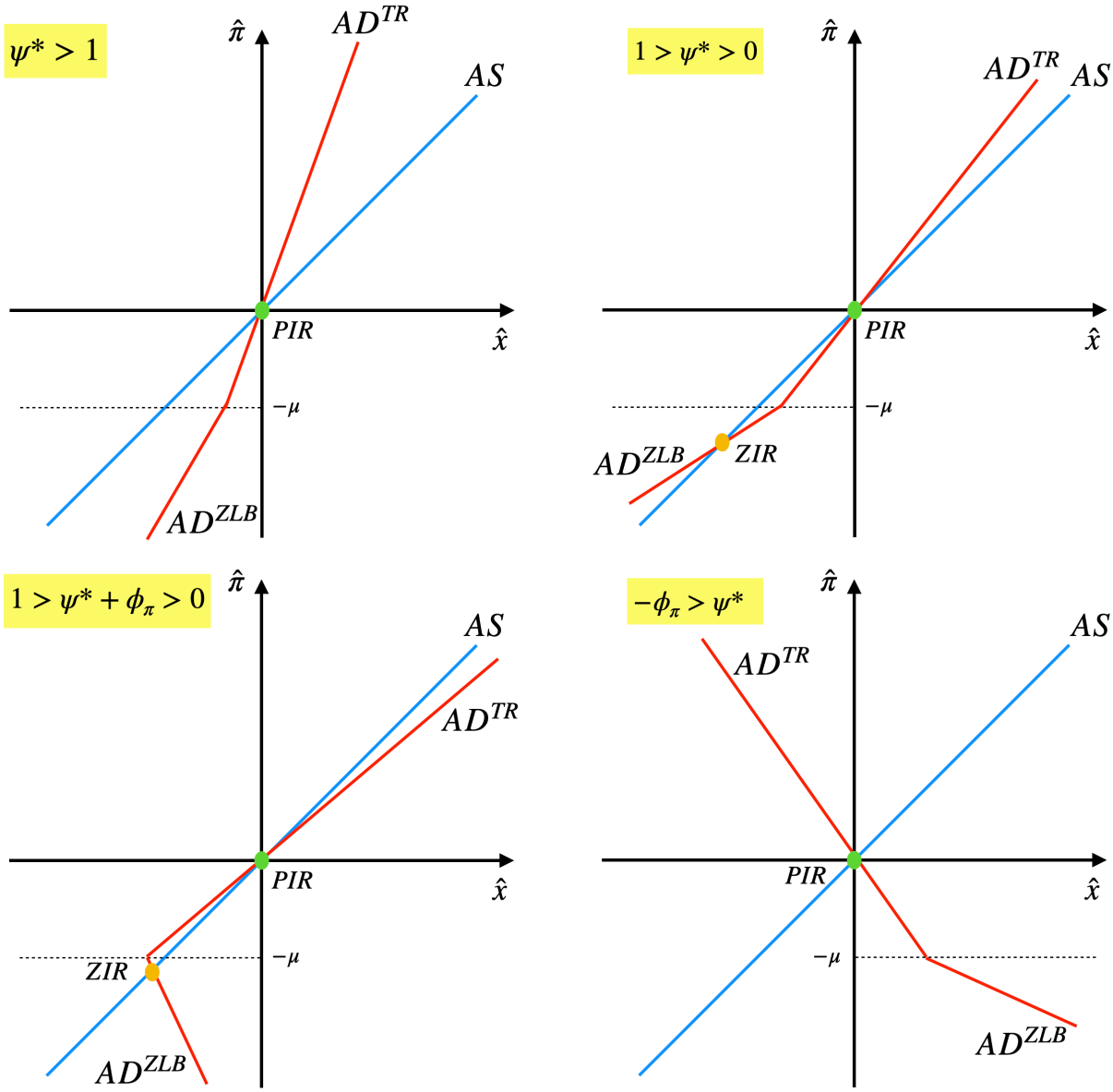
Note: Blue circles denote regions where coherency and completeness conditions are satisfied. Red triangles denote region where the model is either incoherent or incomplete.

Figure 9: Coherency and Completeness Region for Simple Inflation Targeting Fiscal Rule (Equation (34))



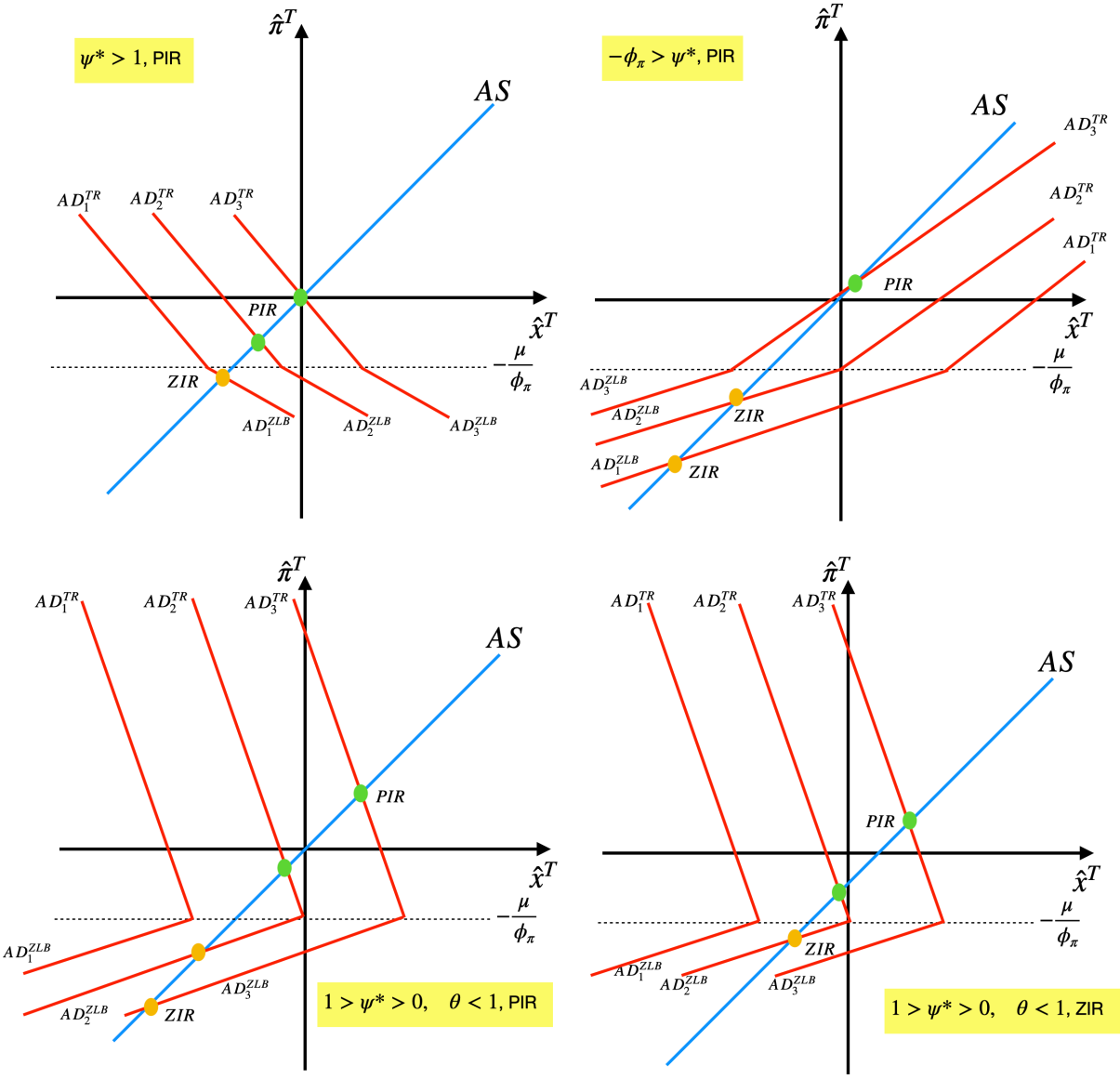
Note: Blue circles denote regions where coherency and completeness conditions are satisfied. Red triangles denotes region where the model is either incoherent or incomplete.

Figure 10: Coherency and Completeness with Active and Passive Simple Fiscal Rules (Absorbing State; $\varepsilon_t = 0$)



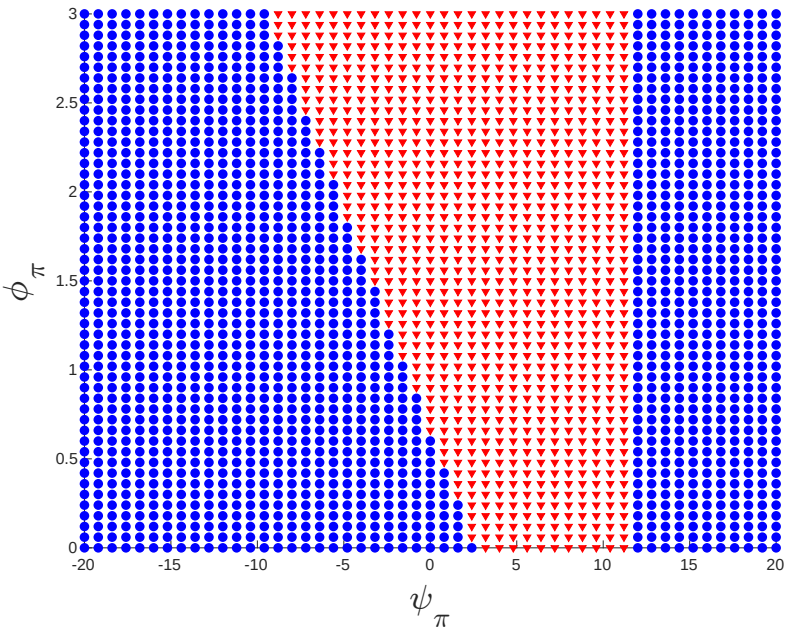
Note: the figure illustrates uniqueness of absorbing state under active fiscal policy (top-left and bottom right). If the fiscal policy is insufficiently aggressive, there are multiple absorbing states (top-right and bottom-left).

Figure 11: Transitory States under Active and Passive Fiscal Policy

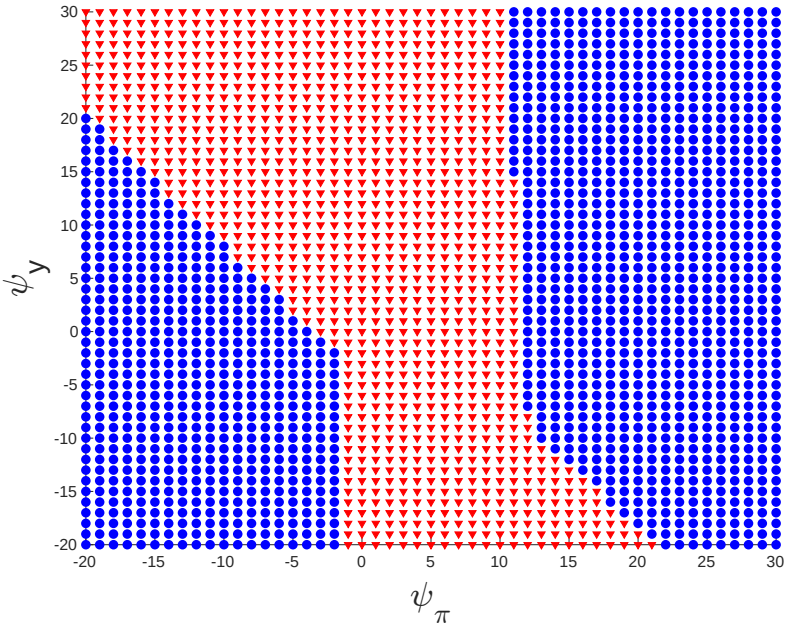


Note: Top row shows transitory state with a PIR absorbing state with active FP. Top left panel shows procyclical FP. Top right shows countercyclical FP. Bottom panels shows passive procyclical FP regime. Passive fiscal policy in general implies non-existence of solution or two solutions as a special case. Active fiscal policy implies existence of unique solution.

Figure 12: Coherency and Completeness Regions for Consumption Tax FP Regimes



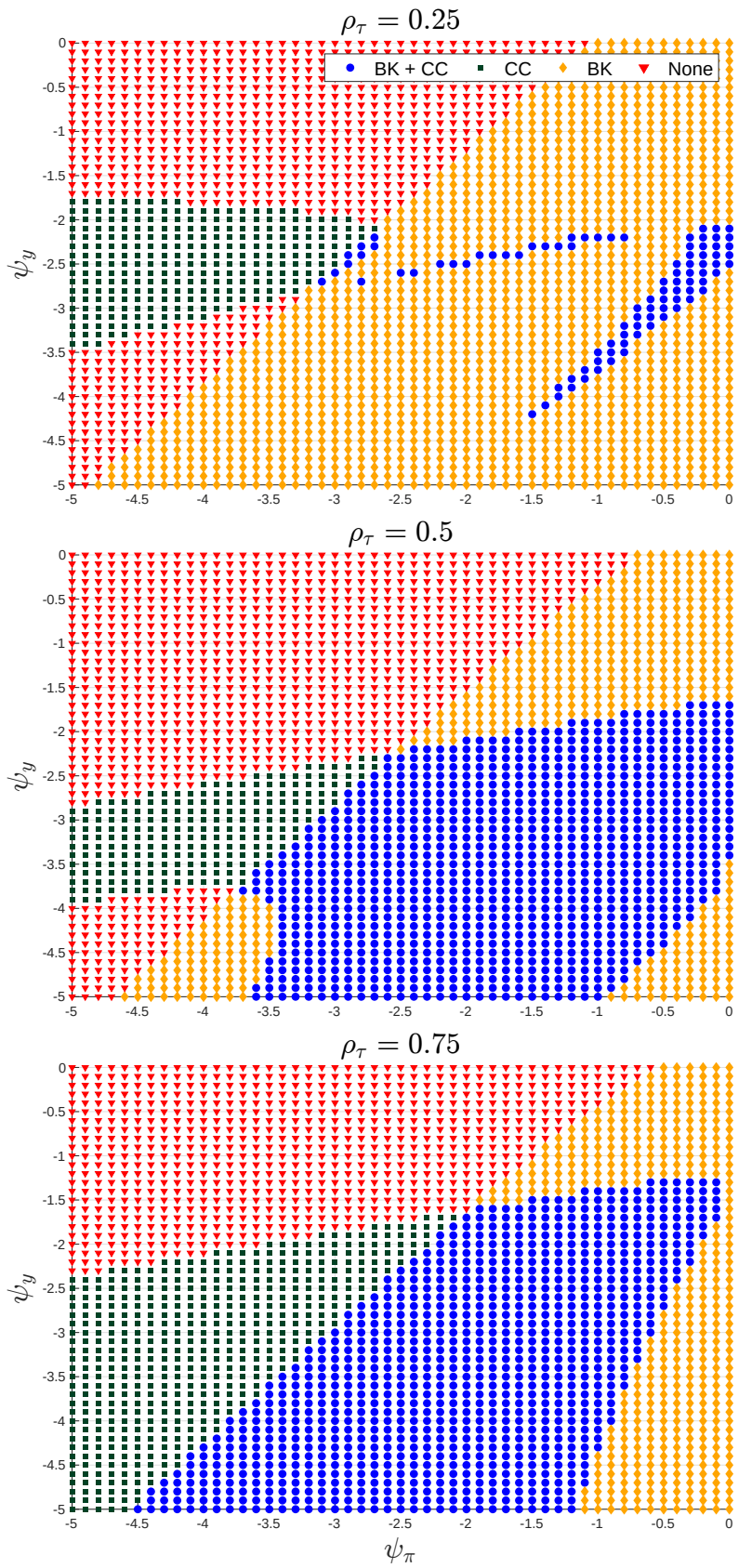
(a) Simple Inflation Targeting FP ($\psi_y = 0$)



(b) Inflation and Output Gap Rule

Note: Blue circles denote regions where coherency and completeness conditions are satisfied. Red triangles denote regions where the model is incoherent or incomplete.

Figure 13: Coherency and Completeness Region with Persistent Fiscal Rule



Note: Figure shows regions where both coherency and completeness and Blanchard-Kahn conditions are satisfied (blue circles) for different values of policy inertia, ρ_τ.

Table 1: Parameter Values and Descriptions

σ	2	Coefficient of relative risk-aversion
φ	2/3	Frisch elasticity of labour supply
β	0.99	Discount factor
τ^c	1/4	Steady-state level of fiscal instrument
κ_y	0.23	Slope of NKPC
Ψ^c	1/4	Coefficient on fiscal instrument
γ	3/4	Calvo probability
ϵ	10	Elasticity of substitution between goods
c	3/4	Fraction of consumption in output
g	1/4	Fraction of government spending in output
ϕ_π	1.5	Weight on inflation, Taylor Rule
ϕ_y	0.2	Weight on output gap, Taylor Rule

A Coherency and Completeness Conditions in a Baseline New Keynesian Model

The baseline New Keynesian model is given by the following three equations:

$$\text{DISE: } \hat{y}_t = \mathbb{E}_t \hat{y}_{t+1} - \frac{1}{\sigma}(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}) + \varepsilon_t, \tag{49}$$

$$\text{NKPC: } \hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa \hat{y}_t + u_t, \tag{50}$$

$$\text{TR: } i_t = \max\left\{-\mu, \phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t\right\}. \tag{51}$$

When the constraint on i_t is binding, the system can be rewritten as follows

$$\begin{pmatrix} 1 & -\kappa \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{\pi}_t \\ \hat{y}_t \end{pmatrix} + \begin{pmatrix} -\beta & 0 \\ -\frac{1}{\sigma} & -1 \end{pmatrix} \begin{pmatrix} \hat{\pi}_{t+1} \\ \hat{y}_{t+1} \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_t \\ \varepsilon_t \end{pmatrix} = 0 \tag{52}$$

Whilst when the constraint is slack the system is given by

$$\begin{pmatrix} 1 & -\kappa \\ \frac{\phi_\pi}{\sigma} & 1 + \frac{\phi_y}{\sigma} \end{pmatrix} \begin{pmatrix} \hat{\pi}_t \\ \hat{y}_t \end{pmatrix} + \begin{pmatrix} -\beta & 0 \\ -\frac{1}{\sigma} & -1 \end{pmatrix} \begin{pmatrix} \hat{\pi}_{t+1} \\ \hat{y}_{t+1} \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_t \\ \varepsilon_t \end{pmatrix} = 0 \tag{53}$$

The model can be cast in the canonical form as in (2), which can be expressed as a mapping $F(\mathbf{Y}) = \lambda(\mathbf{X})$. To check whether the model satisfies the CC condition, it is sufficient to check the invertability of $F(\cdot)$. The function $F(\mathbf{Y})$ can be expressed as in (4) and (8). The relevant coefficients are given by:

$$\boldsymbol{A}_0 = \begin{pmatrix} 1 & -\kappa \\ 0 & 1 \end{pmatrix}, \quad \boldsymbol{A}_1 = \begin{pmatrix} 1 & -\kappa \\ \frac{\phi_\pi}{\sigma} & 1 + \frac{\phi_y}{\sigma} \end{pmatrix}, \quad \boldsymbol{B}_0 = \boldsymbol{B}_1 = \begin{pmatrix} -\beta & 0 \\ -\frac{1}{\sigma} & -1 \end{pmatrix}.$$

Observe that assuming $p = q = 1$ the determinants of $\boldsymbol{\mathcal{A}}_{J_1}, \boldsymbol{\mathcal{A}}_{J_4}$ are

$$|\boldsymbol{\mathcal{A}}_{J_1}| = \det \begin{pmatrix} 1-\beta & -\kappa \\ \frac{\phi_\pi - 1}{\sigma} & \frac{\phi_y}{\sigma} \end{pmatrix} = \frac{(1-\beta)\phi_y + \kappa(\phi_\pi - 1)}{\sigma} > 0, \tag{54}$$

$$|\boldsymbol{\mathcal{A}}_{J_4}| = \det \begin{pmatrix} 1-\beta & -\kappa \\ -\frac{1}{\sigma} & 0 \end{pmatrix} = -\frac{\kappa}{\sigma} < 0. \tag{55}$$

We observe that the signs of $|\boldsymbol{\mathcal{A}}_{J_1}|$ and $|\boldsymbol{\mathcal{A}}_{J_4}|$ differ, which implies that the model is not generally coherent under an active TR with $\phi_\pi > 1$ and $\phi_y \geq 0$.

A.1 Analytical Derivation of CC Conditions

To attain an analytical expression for the CC conditions for the baseline New Keynesian model, we first look for a solution of the form $\hat{\pi}_t = f_\pi(\varepsilon_t)$ and $\hat{y}_t = f_y(\varepsilon_t)$. Let ε_t denote the vector k states of the shock and similarly for the solutions π and y. Denote $\boldsymbol{K}$ as the transition kernel of the Markov chain for ε_t. Then, with some abuse of notation, define $\mathbb{E}_t\hat{\pi}_{t+1} = \boldsymbol{K\pi}$ and $\mathbb{E}_t\hat{y}_{t+1} = \boldsymbol{Ky}$, then rewrite the model equations (1a)-(1c) as the following:

$$\begin{aligned} \boldsymbol{Ky} &= \boldsymbol{y} + \frac{1}{\sigma}(\boldsymbol{i} - \boldsymbol{K\pi}) - \boldsymbol{\varepsilon}, \\ \boldsymbol{\pi} &= \beta\boldsymbol{K\pi} + \kappa\boldsymbol{y}, \\ \boldsymbol{i} &= \max\left\{-\mu\boldsymbol{\iota}, \phi_\pi\boldsymbol{\pi}\right\}, \end{aligned}$$

where for ease of exposition we have assumed that $u_t = 0$ and $\phi_y = 0$. To clarify the notation: symbols in bold are either vectors or matrices, and $\boldsymbol{\iota}$ is a k-length unit vector.

The DISE can be written as:

$$(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{y} = -\frac{1}{\sigma}\left(\max\left\{-\mu\boldsymbol{\iota}, \phi_\pi\boldsymbol{\pi}\right\} - \boldsymbol{K}\boldsymbol{\pi}\right) + \boldsymbol{\varepsilon}.$$

Then, premultiply the NKPC by $(\boldsymbol{I}-\boldsymbol{K})$ to get

$$(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{\pi} = \kappa(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{y} + \beta(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{K}\boldsymbol{\pi}.$$

Then, substitute the expression for $(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{y}$ from the DISE and do some slight rearranging to write:

$$\left[\boldsymbol{I}-\boldsymbol{K}-\frac{\kappa}{\sigma}\boldsymbol{K}-\beta(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{K}\right]\boldsymbol{\pi} = -\frac{\kappa}{\sigma}\max\left\{-\mu\boldsymbol{\iota}, \phi_\pi\boldsymbol{\pi}\right\} + \kappa\boldsymbol{\varepsilon}.$$

Continue rearranging this expression:

$$\begin{aligned}
\left[\boldsymbol{I}-\boldsymbol{K}-\frac{\kappa}{\sigma}\boldsymbol{K}-\beta(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{K}\right]\boldsymbol{\pi} &= \frac{\kappa\mu}{\sigma}\boldsymbol{\iota} - \frac{\kappa}{\sigma}\max\left\{\boldsymbol{0}, \phi_\pi\boldsymbol{\pi}+\mu\boldsymbol{\iota}\right\} + \kappa\boldsymbol{\varepsilon} \\
\left[\boldsymbol{I}-\boldsymbol{K}-\frac{\kappa}{\sigma}\boldsymbol{K}-\beta(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{K}\right]\boldsymbol{\pi} &= (1-1)\left[\boldsymbol{I}-\boldsymbol{K}-\frac{\kappa}{\sigma}\boldsymbol{K}-\beta(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{K}\right]\frac{\mu}{\phi_\pi}\boldsymbol{\iota} \\
&\quad + \frac{\kappa\mu}{\sigma}\boldsymbol{\iota} - \frac{\kappa}{\sigma}\max\left\{\boldsymbol{0}, \phi_\pi\left(\boldsymbol{\pi}+\frac{\mu}{\phi_\pi}\boldsymbol{\iota}\right)\right\} + \kappa\boldsymbol{\varepsilon} \\
\left[\boldsymbol{I}-\boldsymbol{K}-\frac{\kappa}{\sigma}\boldsymbol{K}-\beta(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{K}\right]\left(\boldsymbol{\pi}+\frac{\mu}{\phi_\pi}\boldsymbol{\iota}\right) &= \left[\boldsymbol{I}-\boldsymbol{K}-\frac{\kappa}{\sigma}\boldsymbol{K}-\beta(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{K}\right]\frac{\mu}{\phi_\pi}\boldsymbol{\iota} \\
&\quad + \frac{\kappa\mu}{\sigma}\boldsymbol{\iota} - \frac{\kappa}{\sigma}\max\left\{\boldsymbol{0}, \phi_\pi\left(\boldsymbol{\pi}+\frac{\mu}{\phi_\pi}\boldsymbol{\iota}\right)\right\} + \kappa\boldsymbol{\varepsilon}.
\end{aligned}$$

But since $\left[(\boldsymbol{I}-\boldsymbol{K}) - a\boldsymbol{K} - b(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{K}\right]\boldsymbol{\iota} = -a\boldsymbol{\iota}$ for generic scalars a and b, we can write:

$$\left[\boldsymbol{I}-\boldsymbol{K}-\frac{\kappa}{\sigma}\boldsymbol{K}-\beta(\boldsymbol{I}-\boldsymbol{K})\boldsymbol{K}\right]\left(\boldsymbol{\pi}+\frac{\mu}{\phi_\pi}\boldsymbol{\iota}\right) = -\frac{\kappa}{\sigma}\frac{\mu}{\phi_\pi}\boldsymbol{\iota} + \frac{\kappa\mu}{\sigma}\boldsymbol{\iota} + \kappa\boldsymbol{\varepsilon}$$

$$
\begin{aligned}
& & & - \max\left\{\mathbf{0}, \frac{\kappa\phi_\pi}{\sigma}\left(\boldsymbol{\pi} + \frac{\mu}{\phi_\pi}\boldsymbol{\iota}\right)\right\} \\
\left[\boldsymbol{I} - \boldsymbol{K} - \frac{\kappa}{\sigma}\boldsymbol{K} - \beta(\boldsymbol{I} - \boldsymbol{K})\boldsymbol{K}\right]\left(\boldsymbol{\pi} + \frac{\mu}{\phi_\pi}\boldsymbol{\iota}\right) & = & & \frac{\kappa\mu}{\sigma}\left(\frac{\phi_\pi - 1}{\phi_\pi}\right)\boldsymbol{\iota} + \kappa\boldsymbol{\varepsilon} \\
& & & - \max\left\{\mathbf{0}, \frac{\kappa\phi_\pi}{\sigma}\left(\boldsymbol{\pi} + \frac{\mu}{\phi_\pi}\boldsymbol{\iota}\right)\right\} \\
\left[\boldsymbol{K} - \boldsymbol{I} + \frac{\kappa}{\sigma}\boldsymbol{K} + \beta(\boldsymbol{I} - \boldsymbol{K})\boldsymbol{K}\right]\left(\boldsymbol{\pi} + \frac{\mu}{\phi_\pi}\boldsymbol{\iota}\right) & = & & \frac{\kappa\mu}{\sigma}\left(\frac{1 - \phi_\pi}{\phi_\pi}\right)\boldsymbol{\iota} - \kappa\boldsymbol{\varepsilon} \\
& & & + \max\left\{\mathbf{0}, \frac{\kappa\phi_\pi}{\sigma}\left(\boldsymbol{\pi} + \frac{\mu}{\phi_\pi}\boldsymbol{\iota}\right)\right\}.
\end{aligned}
$$

The above system can be generically written as

$$
\boldsymbol{B}\boldsymbol{v} = \boldsymbol{b} + \max\left\{\mathbf{0}, \boldsymbol{D}\boldsymbol{v}\right\}, \tag{56}
$$

where

$$
\begin{aligned}
\boldsymbol{B} &= \boldsymbol{K} - \boldsymbol{I} + \frac{\kappa}{\sigma}\boldsymbol{K} + \beta(\boldsymbol{I} - \boldsymbol{K})\boldsymbol{K}, \\
\boldsymbol{v} &= \boldsymbol{\pi} + \frac{\mu}{\phi_\pi}\boldsymbol{\iota}, \\
\boldsymbol{b} &= \frac{\kappa\mu}{\sigma}\left(\frac{1 - \phi_\pi}{\phi_\pi}\right)\boldsymbol{\iota} - \kappa\boldsymbol{\varepsilon}, \\
\boldsymbol{D} &= \frac{\kappa\phi_\pi}{\sigma}\boldsymbol{I}.
\end{aligned}
$$

The CC conditions can be analytical derived in the case that $k = 2$. We thus have a piecewise linear system in four orthants that can be written as:

$$
\begin{cases}
\mathbb{R}_1 & = \{(v_1, v_2) : v_1 \geq 0, v_2 \geq 0\}, \\
\boldsymbol{A}_1 & = \boldsymbol{B} - \boldsymbol{D},
\end{cases}
$$

$$
\begin{cases} \mathbb{R}_2 & = \{(v_1, v_2) : v_1 \geq 0, v_2 < 0\}, \\ \boldsymbol{A}_2 & = \boldsymbol{B} - \begin{pmatrix} \frac{\kappa\phi_\pi}{\sigma} & 0 \\ 0 & 0 \end{pmatrix}, \end{cases}
$$

$$
\begin{cases} \mathbb{R}_3 & = \{(v_1, v_2) : v_1 < 0, v_2 < 0\}, \\ \boldsymbol{A}_3 & = \boldsymbol{B}, \end{cases}
$$

$$
\begin{cases} \mathbb{R}_4 & = \{(v_1, v_2) : v_1 < 0, v_2 \geq 0\}, \\ \boldsymbol{A}_4 & = \boldsymbol{B} - \begin{pmatrix} 0 & 0 \\ 0 & \frac{\kappa\phi_\pi}{\sigma} \end{pmatrix}. \end{cases}
$$

Let $\boldsymbol{K}$ be defined as:

$$
\boldsymbol{K} = \begin{pmatrix} p & 1-p \\ 1-q & q \end{pmatrix}.
$$

Theorem 1 of Gourieroux, Laffont, and Monfort (1980) states that the system of equations (56) is coherent and complete if and only if all the determinants of the matrices below have the same sign:

$$
\begin{aligned}
\det \boldsymbol{A}_1 &= \frac{\kappa(1-\phi_\pi)}{\sigma}\left[(p+q-1)\frac{\kappa}{\sigma} + a(p+q-2) - \frac{\kappa\phi_\pi}{\sigma}\right], \\
\det \boldsymbol{A}_2 &= \frac{\kappa}{\sigma}\left[a(p-2) + (a\sigma - \kappa\phi_\pi)\frac{q}{\sigma} - a(q-1)\phi_\pi + (p+q-1)\frac{\kappa}{\sigma}\right], \\
\det \boldsymbol{A}_3 &= \frac{\kappa}{\sigma}\left[(p+q-1)\frac{\kappa}{\sigma} + a(p+q-2)\right], \\
\det \boldsymbol{A}_4 &= \frac{\kappa}{\sigma}\left[a(q-2) + (a\sigma - \kappa\phi_\pi)\frac{p}{\sigma} - a(p-1)\phi_\pi + (p+q-1)\frac{\kappa}{\sigma}\right],
\end{aligned}
$$

where $a = (1 - \beta(p+q-1))$. Alternatively, we could write the above matrices more

compactly as:

$$\det \boldsymbol{A}_1 = (1-\phi_\pi)\left(\det \boldsymbol{B} - \frac{\kappa^2\phi_\pi}{\sigma^2}\right), \tag{57}$$

$$\det \boldsymbol{A}_2 = \det \boldsymbol{B} + \frac{\kappa\phi_\pi}{\sigma}\left[(1-q)(1+\beta(1-p-q)) - \frac{\kappa}{\sigma}q\right], \tag{58}$$

$$\det \boldsymbol{A}_3 = \det \boldsymbol{B}, \tag{59}$$

$$\det \boldsymbol{A}_4 = \det \boldsymbol{B} + \frac{\kappa\phi_\pi}{\sigma}\left[(1-p)(1+\beta(1-p-q)) - \frac{\kappa}{\sigma}p\right]. \tag{60}$$

It is evident that the CC conditions will crucially depend on the sign of $\det \boldsymbol{B}$, which we can write as:

$$\det \boldsymbol{B} = \frac{\kappa}{\sigma}\left[(p+q-1)\frac{\kappa}{\sigma} - (1-\beta(p+q-1))(2-p-q)\right]. \tag{61}$$

Observing this quantity, we know that $\frac{\kappa}{\sigma} > 0$ and that $(1-\beta(p+q-1))(2-p-q) > 0$. Thus, we need to check the relative value of $(p+q-1)\frac{\kappa}{\sigma}$ to see if $\det \boldsymbol{B}$ is greater or less than zero. We thus need to check two cases:

Case 1: $(p+q-1)\frac{\kappa}{\sigma} > (1-\beta(p+q-1))(2-p-q)$. The RHS of the inequality is always greater than 0, thus implying:

$$p+q-1 > 0 \implies \det \boldsymbol{B} > 0.$$

Given this, what does it mean for $\det \boldsymbol{A}_i$ for $i = \{1,2,4\}$? First, rearrange the quantities for $\det \boldsymbol{A}_1 > 0$ in Equation (57) to write:

$$\phi_\pi < \frac{\sigma^2}{\kappa^2}\det \boldsymbol{B},$$

as in $\det \boldsymbol{A}_1 > 0$ if and only if

$$
\begin{aligned}
0 &< (1-\phi_\pi)(\det \boldsymbol{B} - \frac{\kappa^2}{\sigma^2}\phi_\pi), \\
\therefore \phi_\pi &< \min\left\{1, \frac{\sigma^2}{\kappa^2}\det \boldsymbol{B}\right\} \Leftrightarrow \phi_\pi > \max\left\{1, \frac{\sigma^2}{\kappa^2}\det \boldsymbol{B}\right\}.
\end{aligned}
$$

For $\det \boldsymbol{A}_2$ we need

$$
\frac{\kappa\phi_\pi}{\sigma}\left[(1-q)(1-\beta(p+q-1)) - \frac{\kappa}{\sigma}q\right] + \det \boldsymbol{B} > 0.
$$

If the term in the square brackets is greater than zero, then $\det \boldsymbol{A}_2 > 0, \forall \phi_\pi \geq 0$, so long as $\det \boldsymbol{B} > 0$ (which was shown above). But what if this bracketed quantity is less than zero? Then, write $\det \boldsymbol{A}_2$ as:

$$
\begin{aligned}
0 < \frac{\kappa\phi_\pi}{\sigma}&\left[\underbrace{(1-q)(1-\beta(p+q-1)) - \frac{\kappa}{\sigma}q}_{<0}\right] \\
&+ \frac{\kappa}{\sigma}\left[(p+q-1)\frac{\kappa}{\sigma} - (1-\beta(p+q-1))(2-p-q)\right],
\end{aligned}
$$

where we can write:

$$
\phi_\pi < \frac{(p+q-1)\frac{\kappa}{\sigma} - (1-\beta(p+q-1))(2-p-q)}{\underbrace{\frac{\kappa}{\sigma}q - (1-q)(1-\beta(p+q-1))}_{>0}} < 1.
$$

Use (A.1) to then write:

$$
\begin{aligned}
\phi_\pi &< \frac{\sigma^2}{\kappa^2}\det \boldsymbol{B} \\
&< p+q-1 - \frac{\sigma}{\kappa}((1-\beta(p+q-1))(2-p-q).
\end{aligned}
$$

A symmetric argument holds for $\det \boldsymbol{A}_4$.

Case 2: $(p+q-1)\frac{\kappa}{\sigma} < (1-\beta(p+q-1))(2-p-q)$ This case now assumes that $\det \boldsymbol{B} < 0$, so we need all the other determinants to be negative too.

For $\det \boldsymbol{A}_1$, clearly for any $0 \leq \phi_\pi < 1$, $\det \boldsymbol{A}_1 < 0$.

$\det \boldsymbol{A}_2$ is negative $\forall \phi_\pi \geq 0$ if and only if

$$(1-q)(1+\beta(1-p-q)) - \frac{\kappa}{\sigma}q < 0.$$

But if

$$(1-q)(1+\beta(1-p-q)) - \frac{\kappa}{\sigma}q > 0,$$

then we have

$$\begin{aligned} 0 &> \det \boldsymbol{B} + \frac{\kappa\phi_\pi}{\sigma}\left[(1-q)(1+\beta(1-p-q)) - \frac{\kappa}{\sigma}q\right] \\ \phi_\pi &< -\frac{\sigma}{\kappa}\frac{\det \boldsymbol{B}}{\left[(1-q)(1+\beta(1-p-q)) - \frac{\kappa}{\sigma}q\right]} < 0. \end{aligned}$$

Denote ϕ_π^* as

$$\phi_\pi^* = p+q-1-\frac{\sigma}{\kappa}(1-\beta(p+q-1))(2-p-q),$$

then we can write the region for which the baseline NK model satisfies the CC conditions as:

$$\begin{cases} \phi_\pi < \phi_\pi^*, & \text{if } \phi_\pi^* > 0, \\ \phi_\pi < 1, & \text{if } \phi_\pi^* < 0. \end{cases}$$

This is (12) in the text.

A.1.1 Derivation for Graphical Representation

First consider the absorbing state when $\varepsilon = 0$, and also when $\phi_y = 0$, $u_t = 0$. From (1a)-(1c), we can write AS as:

$$\hat{\pi} = \frac{\kappa}{1-\beta}\hat{y}. \tag{62}$$

Meanwhile, from the DISE we have:

$$\hat{\pi} = \begin{cases} \phi_\pi \hat{\pi} & \text{TR}, \\ -\mu & \text{ZLB}. \end{cases}$$

Substituting AS into the above expression gives AD:

$$\hat{\pi} = \begin{cases} \frac{\kappa\phi_\pi}{1-\beta}\hat{y} & AD^{TR}, \\ -\mu & AD^{ZLB}. \end{cases} \tag{63}$$

We plot AS and AD in Figure 1.

Next, consider the transitory state when $\varepsilon = \frac{\rho}{\sigma}\hat{r}^T \neq 0$.

PIR absorbing state. At time t the economy is in a transitory state. With probability p the economy remains in the transitory state, and with $(1-p)$ the economy moves to the PIR absorbing state. From (1b), we can write:

$$\hat{\pi}^T = \kappa\hat{y}^T + p\beta\hat{\pi}^T,$$

where the second term on the RHS comes from the fact that in period $t+1$ you may be in a transitory state where $\pi \neq 0$. Thus, AS is:

$$\hat{\pi}^T = \frac{\kappa}{1-p\beta}\hat{y}^T. \tag{64}$$

For AD, begin by writing the DISE as:

$$\hat{y}^T = p\hat{y}^T - \frac{1}{\sigma}(\hat{i} - p\hat{\pi}^T) + \varepsilon.$$

Rearrange and substitute in (1c) and $\varepsilon = \frac{p}{\sigma}\hat{r}^T$ to get AD:

$$\hat{\pi}^T = \begin{cases} \frac{\sigma(1-p)}{p-\phi_\pi^T}\hat{y}^T - \frac{p}{p-\phi_\pi}\hat{r}^T & AD^{TR} \text{ for } \hat{\pi}^T \geq -\frac{\mu}{\phi_\pi}, \\ \frac{\sigma(1-p)}{p}\hat{y}^T - \frac{\mu}{p} - \hat{r}^T & AD^{ZLB} \text{ for } \hat{\pi}^T \leq -\frac{\mu}{\phi_\pi}. \end{cases} \tag{65}$$

ZIR absorbing state. Here in period t the economy is in a transitory state. With probably p the economy can remain in a transitory state, and with $(1-p)$ it can move to a ZIR absorbing state. Therefore, from (1b), AS can be written as:

$$\begin{aligned} \hat{\pi}^T &= \beta(p\hat{\pi}^T + (1-p)(-\mu)) + \kappa\hat{y}^T \\ &= \frac{\kappa}{1-p\beta}\hat{y}^T - \frac{\beta(1-p)}{1-p\beta}\mu. \end{aligned} \tag{66}$$

To find AD, first begin by writing the DISE as:

$$\hat{y}^T = \left[p\hat{y}^T + (1-p)\left(\frac{-\mu(1-\beta)}{\kappa}\right)\right] - \frac{1}{\sigma}\left[\hat{i} - \left(p\hat{\pi}^T + (1-p)(-\mu)\right)\right] + \varepsilon,$$

then substitute in (1c) and the ε to get AD:

$$\hat{\pi}^T = \begin{cases} \frac{\sigma(1-p)}{p-\phi_\pi}\hat{y}^T + \frac{1-p}{p-\phi_\pi}\left[\frac{(1-\beta)}{\kappa}+1\right]\mu - \frac{p}{p-\phi_\pi}\hat{r}^T & AD^{TR} \text{ for } \pi^T \geq -\frac{\mu}{\phi_\pi}, \\ \frac{\sigma(1-p)}{p}\hat{y}^T + \frac{1-p}{p}\left[\frac{(1-\beta)\sigma}{\kappa}+1\right]\mu - \frac{\mu}{p} - \hat{r}^T & AD^{ZLB} \text{ for } \hat{\pi}^T \leq -\frac{\mu}{\phi_\pi}. \end{cases} \tag{67}$$

To find θ simply divide the slop of AD^{TR} by the slope of AS:

$$\begin{aligned} \theta &= \frac{\partial AD^{ZLB}/\partial \hat{y}^T}{\partial AS/\partial \hat{y}^T} \\ &= \frac{\frac{1}{p}-\sigma}{\frac{\kappa}{1-p\beta}} \\ &= \frac{\sigma(1-p)(1-p\beta)}{p\kappa}. \end{aligned}$$

We plot AD and AS for when the economy is in the transitory state and with PIR and ZIR absorbing states for $\theta > 1$ and $\theta < 1$ in Figure 2.

A.1.2 Proof of Support Restrictions

For the case of $\theta > 1$, the two solutions imply that

$$\frac{1}{r\pi^*} \leq 1$$

is a necessary support restrictions. Why? If $r\pi^* < 1$, then either the gross real interest rate, the gross target rate of inflation, or their product is less than one. But this cannot be the case as we define $\mu = \log(r\pi^*)$.

For the case of $\theta < 1$, we need further support restrictions to ensure coherency. This can be found by finding the point at which AD and AS intersect at the kink of AD. The case with a PIR absorbing state is analytically more tractable, so we focus on that.

$AD = AS$ when $\pi^T = -\frac{\mu}{\phi_\pi}$, and when we wish to find shock size $\hat{r}^T = \bar{r}^T$ such that the equations have a solution for all $-\hat{r}^T \leq -\bar{r}^T$. Hence, the cutoff can be found by setting AS and AD^{TR} equal:

$$\begin{aligned} AS: \hat{y}^T &= \frac{(1-p\beta)}{\kappa}\hat{\pi}^T \\ AD^{TR}\, \hat{y}^T &= \frac{(\phi_\pi - p}{\sigma(1-p)}\hat{\pi}^T + \frac{p}{\sigma(1-p)}\hat{r}^T. \end{aligned}$$

Substitute in $\hat{\pi}^T = -\frac{\mu}{\phi_\pi}$ and rearrange to get:

$$-\hat{r}^T = \mu\left(\frac{\theta}{\phi_\pi} + \frac{\phi_\pi - p}{p\phi_\pi}\right). \tag{68}$$

B A New Keynesian Model with Fiscal Policy

Households. The economy is populated with households indexed with i on a continuum of measure one. The households gain utility from consumption, dislike labour, and have access to one-period risk free bonds. The optimisation problem of the households is thus:

$$\max_{\{C_t, L_t, B_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{L_t^{1+\varphi}}{1+\varphi}\right) Z_t,$$

subject to the [nominal] period budget constraint is given by

$$(1-\tau_t^c)P_t C_t + B_t = (1-\tau_t^w)W_t L_t + R_{t-1}B_{t-1} + T_t^h,$$

where C_t is consumption, L_t is labour supply, B_t denotes bonds, R_t is nominal interest rate, P_t is the price level, τ_t^c is the consumption tax rate, τ_t^w is the wage tax rate, and T_t^h are lump-sum taxes.

The consumption bundle C_t consists of a continuum of differentiated goods, and is bundled by a CES aggregator of the form:

$$C_t = \left[\int_0^1 C_t(j)^{\frac{\epsilon-1}{\epsilon}} \, dj \right]^{\frac{\epsilon}{\epsilon-1}}.$$

The utility maximisation problem of the household results in the following intertemporal Euler equation:

$$\beta \mathbb{E}_t \frac{R_t}{\Pi_{t+1}} \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{Z_{t+1}}{Z_t} = \mathbb{E}_t \frac{1-\tau^c_{t+1}}{1-\tau^c_t}.$$

The labour supply condition gives the following intratemporal Euler equation:

$$\frac{1-\tau^w_t}{1-\tau^c_t} w_t C_t^{-\sigma} = L_t^{\varphi}.$$

The intratemporal household problem of choosing a consumption bundle results in the following demand for good j:

$$C_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\epsilon} C_t.$$

Production. Producers use labour as an input to produce differentiated consumption goods according to the following production technology:

$$Y_t(j) = A_t L_t(j).$$

The price-setting problem of an individual firm j follows Rotemberg (1982) where firm j maximises the discounted value of profits,

$$\max_{\{P_t(i)\}} \mathbb{E}_t \sum_{T=t}^{\infty} Q_{t,T} \left[(1-\tau_t^s) P_t(j) Y_{t,T}(j) - w_T L_T(j) - \frac{\Phi}{2} \left(\frac{P_{t,T}(j)}{P_{t-1,T}(j)} - 1 \right)^2 Y_{t,T} \right],$$

subject to:

$$Y_{t,T}(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\epsilon} Y_t,$$

where Φ denotes a price adjustment cost parameter for the firms.[18] $Y_{t,T}(j)$ denotes demand at time T conditional on the price unchanged since period t. The firm maximises infinite discounted stream of profits, with revenues given by the first term and costs given by the second term. The revenues of the firm are taxed with tax level denoted by τ_t^s. Households own firms, thus their revenues are discounted with the households' discount factor, $Q_{t,T}$:

$$Q_{t,T} = \beta \frac{P_t}{P_T} \left(\frac{C_T}{C_t} \right)^{-\sigma} \frac{Z_T}{Z_t}.$$

The solution to the firm problem results in the following equation for inflation:[19]

$$\Pi_t(\Pi_t - 1) = \frac{1}{\kappa} \left[\epsilon mc_t + 1 - \epsilon + \tau_t^s \epsilon - \tau_t^s \right] + \mathbb{E}_t \left[Q_{t,t+1} (\Pi_{t+1} - 1) \Pi_{t+1} \frac{Y_{t+1}}{Y_t} \right].$$

18. We calibrate Φ to the following:

$$\Phi = \frac{\epsilon\gamma}{(1-\gamma)(1-\beta\gamma)},$$

where γ is the probability of firm j being unable to optimally adjust its price in any given period as in a model with Calvo (1983) pricing.

19. Gross inflation is defined as $\Pi_t = P_t / P_{t-1}$

Monetary authority. The monetary authority uses the [gross] nominal interest rate, R_t, as its policy instrument and sets it according to a TR of the form:

$$\frac{R_t}{\bar{R}} = \max\left\{1, \left(\frac{\Pi_t}{\pi^*}\right)^{\phi_\pi} \left(\frac{Y_t}{\bar{Y}}\right)^{\phi_y}\right\},$$

where ϕ_π and ϕ_y is the degree of reaction to contemporaneous inflation and the output deviations, respectively.

Fiscal authority. The nominal flow budget constraint for the government is

$$\tau_t^c C_t + \tau_t^s Y_t + \tau_t^w w_t L_t + T_t = G_t. \tag{69}$$

We make clear which taxes are enabled or disabled in each section of the paper, as we explore different tax regimes. An additional equation – a rule fiscal policy rule – is needed to close the model. In what follows we explore different specifications of such rules.

Market clearing. Markets clear, hence all output is consumed or used for government expenditure,

$$Y_t = C_t + G_t + \frac{\Phi}{2}\left(\Pi_t - 1\right)^2 Y_t.$$

Note that as we assume Rotemberg adjustment costs, the natural level of output, Y_t^n, coincides with headline output when $\Phi = 0$.

B.1 Log Linearised Equilibrium Conditions

Log linearising the non-linear model equations about a non-inflation deterministic steady state yields the following: Intertemporal Euler equation:[20]

$$\hat{c}_t = \mathbb{E}_t \hat{c}_{t+1} - \frac{1}{\sigma}\left(\hat{i}_t + \mathbb{E}_t\left[\Delta \hat{z}_{t+1} + \Psi^c \Delta \hat{\tau}^c_{t+1} - \hat{\pi}_{t+1}\right]\right). \tag{70}$$

Labour supply condition:[21]

$$\hat{w}_t = \sigma \hat{c}_t + \varphi \hat{l}_t + \hat{\tau}^w_t \Psi^w - \Psi^c \hat{\tau}^c_t. \tag{71}$$

Output:

$$\hat{y}_t = \hat{a}_t + \hat{l}_t. \tag{72}$$

Inflation:

$$\hat{\pi}_t = \frac{1}{\Phi}(\epsilon \hat{mc}_t - \hat{\tau}^s_t) + \beta \mathbb{E}_t \hat{\pi}_{t+1}. \tag{73}$$

Marginal cost:

$$\hat{mc}_t = \hat{w}_t - \hat{a}_t. \tag{74}$$

Taylor Rule:

$$i_t = \max\left\{-\mu, \phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t\right\}. \tag{75}$$

Government budget constraint

$$g\hat{g}_t = \frac{T}{Y}\hat{t}_t + \tau^c c(\hat{\tau}^c_t + \hat{c}_t) + \tau^s(\hat{\tau}^s_t + \hat{y}_t) + \tau^w \frac{wL}{Y}(\hat{\tau}^w_t + \hat{w}_t + \hat{l}_t).$$

20. We define $\Psi^c = \frac{\bar{\tau}^c}{1-\bar{\tau}^c}$ and $\Delta\hat{\tau}^c_{t+1} = \hat{\tau}^c_{t+1} - \hat{\tau}^c_t$.
21. We define $\Psi^w = \frac{\bar{\tau}^w}{1-\bar{\tau}^w}$.

Aggregate resource constraint:

$$\hat{y}_t = c\hat{c}_t + g\hat{g}_t. \tag{76}$$

To get an expression for $\hat{y}_t^n$ in terms of $\hat{a}_t$, start by noting that wages are equal to the marginal product of labour:

$$\hat{w}_t^n = \hat{a}_t, \tag{77}$$

and then combine with (71) to write:

$$\hat{a}_t = \sigma\hat{c}_t^n + \varphi\hat{l}_t^n.$$

Note that we assume that in the natural allocation, taxes and government spending do not fluctuate. Then use (76) to substitute in for $\hat{c}_t^n$, and use (72) to substitute in for $\hat{l}_t^n$:

$$\hat{a}_t = \frac{\sigma}{c}\hat{y}_t^n + \varphi(\hat{y}_t^n - \hat{a}_t).$$

Rearrange the above to write:

$$\hat{y}_t^n = \psi_{ya}\hat{a}_t, \tag{78}$$

where $\psi_{ya} = \frac{1+\varphi}{\varphi+\frac{\sigma}{c}}$.

Then, using (70) and (76), the DISE is given by

$$\hat{y}_t = \mathbb{E}_t\hat{y}_{t+1} - \frac{c}{\sigma}(\hat{i}_t - \mathbb{E}_t\hat{\pi}_{t+1} + \mathbb{E}_t\Delta\hat{z}_{t+1} + \mathbb{E}_t\Delta\hat{\tau}_{t+1}^c) - g\mathbb{E}_t\Delta\hat{g}_{t+1}.$$

We now turn to the NKPC. From (74), (71), and (76), marginal cost is given by

$$\hat{mc}_t = \frac{\sigma}{c}(\hat{y}_t - g\hat{g}_t) + \varphi\hat{y}_t + \hat{\tau}_t^w\Psi^w - \Psi^c\hat{\tau}_t^c - \hat{a}_t(1+\varphi).$$

Plug this into the relationship for inflation implied by Rotemberg pricing (73)

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \frac{\epsilon}{\Phi}\left(\frac{\sigma}{c}(\hat{y}_t - g\hat{g}_t) + \varphi \hat{y}_t + \Psi^w \hat{\tau}_t^w - \Psi^c \hat{\tau}_t^c - \frac{1}{\epsilon}\tau_t^s - (1+\varphi)\hat{a}_t\right),$$

to then yield the NKPC:

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_y \hat{y}_t + \frac{\epsilon}{\Phi}\left(\Psi^w \hat{\tau}_t^w - \Psi^c \hat{\tau}_t^c - \frac{1}{\epsilon}\hat{\tau}_t^s - \frac{\sigma}{c}\hat{g}_t - (1+\varphi)\hat{a}_t\right),$$

where $\kappa_y = \frac{\epsilon}{\Phi}(\frac{\sigma}{c} + \varphi)$.

NKPC can be rewritten in terms of output gap, $\hat{x}_t = \hat{y}_t - \hat{y}_t^n$ as follows

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_y \hat{x}_t + \frac{\epsilon}{\Phi}\left(\Psi^w \hat{\tau}_t^w - \Psi^c \hat{\tau}_t^c - \frac{1}{\epsilon}\hat{\tau}_t^s - \frac{\sigma}{c}\hat{g}_t\right).$$

The model is thus given by

$$\hat{x}_t = \mathbb{E}_t \hat{x}_{t+1} - \frac{c}{\sigma}(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1} - \hat{r}_t^n), \tag{79}$$

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_y \hat{x}_t + \frac{\epsilon}{\Phi}\left(\Psi^w \hat{\tau}_t^w - \Psi^c \hat{\tau}_t^c - \frac{1}{\epsilon}\hat{\tau}_t^s - \frac{\sigma}{c}\hat{g}_t\right), \tag{80}$$

$$\hat{i}_t = \max\{-\mu, \phi_\pi \hat{\pi}_t + \phi_y \hat{x}_t\}, \tag{81}$$

$$g\hat{g}_t = \frac{T}{Y}\hat{\tau}_t + \tau^c c(\hat{\tau}_t^c + \hat{c}_t) + \tau^s(\hat{\tau}_t^s + \hat{x}_t + \hat{y}_t^n) + \tau^w(\hat{\tau}_t^w + \hat{w}_t + \hat{l}_t), \tag{82}$$

$$\hat{r}_t^n = \frac{\sigma \psi_{ya}}{c}\Delta \mathbb{E}_t \hat{a}_t - \mathbb{E}_t \Delta \hat{\tau}_{t+1}^c - \frac{\sigma g}{c}\mathbb{E}_t \Delta \hat{g}_{t+1} - \frac{\sigma}{c}\mathbb{E}_t \Delta \hat{z}_{t+1}, \tag{83}$$

which are equations (19)-(20) in the main text.

As mentioned, in order to close the model, tax rules and fiscal policy rules for $\hat{g}_t$, is required. In the text and subsequent appendix chapters, we describe the set of rules and assumptions we make to close the model.

B.2 Canonical form coefficients, Propositon 1

$$\boldsymbol{A}_1 = \begin{pmatrix} -c\sigma^{-1}\phi_\pi + g\psi_\pi & -1 + g\psi_y - c\sigma^{-1}\phi_y \\ -1 - \kappa_g\psi_\pi & \kappa_y - \kappa_g\psi_y \end{pmatrix} \tag{84}$$

$$\boldsymbol{A}_0 = \begin{pmatrix} g\psi_\pi & -1 + g\psi_y \\ -1 - \kappa_g\psi_\pi & \kappa_y - \kappa_g\psi_y \end{pmatrix} \tag{85}$$

$$\boldsymbol{B}_0 = \boldsymbol{B}_1 = \begin{pmatrix} c\sigma^{-1} - g\psi_\pi & 1 - g\psi_y \\ \beta & 0 \end{pmatrix} \tag{86}$$

B.3 Proof of Proposition 4

Use the NK-FP system in (19) with the following government spending rule:

$$\mathbb{E}_t \Delta\widehat{GR}_{t+1} = \psi_\pi \hat{\pi}_t + \psi_y \hat{x}_t.$$

The constraint on $\hat{i}_t$ can be either binding (ZIR) or slack (PIR). Substitute and rearrange the above system of equations to get the following:

$$\hat{x}_t = \begin{cases} \mathbb{E}_t \hat{x}_{t+1} - \frac{c}{\sigma}\left(\phi_\pi \hat{\pi}_t + \phi_y \hat{x}_t - \mathbb{E}_t \hat{\pi}_{t+1}\right) - g\psi_\pi \pi_t + \varepsilon_t, \\ \mathbb{E}_t \hat{x}_{t+1} - \frac{c}{\sigma}\left(-\mu - \mathbb{E}_t \hat{\pi}_{t+1}\right) - g\psi_\pi \pi_t + \varepsilon_t, \end{cases}$$

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_y \hat{x}_t.$$

The vector of exogenous disturbances is denoted by $\boldsymbol{X}_t = \begin{pmatrix} \varepsilon_t & 0 \end{pmatrix}^\top$ and is assumed to follow a two-state first-order Markov process with transition kernel $\boldsymbol{K}$ defined as

$$\boldsymbol{K} = \begin{pmatrix} p & 1-p \\ 1-q & q \end{pmatrix},$$

where $p, q \in [0,1]$ are transition probabilities. This allows to write the model in the canonical form as in (2), and can be cast in the form $F(\mathbf{Y}) = \lambda(\mathbf{X})$. Following Gourieroux, Laffont, and Monfort (1980), it is sufficient to check that the mapping $F(\mathbf{X})$ is invertible for model coherency. The mapping is as in (4) and (8). The relevant coefficient matrices are given by

$$\boldsymbol{A}_1 = \begin{pmatrix} 1 + \frac{c}{\sigma}\phi_y + g\psi_y & \frac{c}{\sigma}\phi_\pi + g\psi_\pi \\ \kappa_y & -1 \end{pmatrix}, \quad \boldsymbol{A}_0 = \begin{pmatrix} 1 + g\psi_y & g\psi_\pi \\ \kappa_y & -1 \end{pmatrix},$$

$$\boldsymbol{B}_0 = \boldsymbol{B}_1 = \begin{pmatrix} -1 & -\frac{c}{\sigma} \\ 0 & \beta \end{pmatrix}.$$

$\boldsymbol{\mathcal{A}}_{J_1}, \boldsymbol{\mathcal{A}}_{J_2}, \boldsymbol{\mathcal{A}}_{J_3}$, and $\boldsymbol{\mathcal{A}}_{J_4}$ are then given by:

$$\boldsymbol{\mathcal{A}}_{J_1} = \boldsymbol{\mathcal{A}}_{J_3} = \begin{pmatrix} 1 + \frac{c}{\sigma}\phi_y + g\psi_y - p - \frac{c}{\sigma}(1-q) & \frac{c}{\sigma}\phi_\pi + g\psi_\pi - 1 + p - \frac{c}{\sigma}(1-p) \\ \kappa_y + \beta(1-q) & \beta q - 1 \end{pmatrix},$$

$$\boldsymbol{\mathcal{A}}_{J_2} = \boldsymbol{\mathcal{A}}_{J_4} = \begin{pmatrix} g\psi_y + 1 - p - \frac{c}{\sigma}(1-q) & g\psi_\pi - 1 + p - \frac{c}{\sigma}(1-p) \\ \kappa_y + \beta(1-q) & \beta q - 1 \end{pmatrix}.$$

Below we show that when $\psi_\pi \to \infty$, $\text{sign}(\boldsymbol{\mathcal{A}}_{J_1}) = \text{sign}(\boldsymbol{\mathcal{A}}_{J_2}) = \text{sign}(\boldsymbol{\mathcal{A}}_{J_3}) = \text{sign}(\boldsymbol{\mathcal{A}}_{J_4})$

and the model thus satisfies the CC conditions. We start with $\boldsymbol{\mathcal{A}}_{J_1}$, determinant of which is given by

$$
\begin{aligned}
|\boldsymbol{\mathcal{A}}_{J_1}| = |\boldsymbol{\mathcal{A}}_{J_3}| = (\beta q - 1)\left[1 + \frac{c}{\sigma}\phi_y + g\psi_y - p - \frac{c}{\sigma}(1-q)\right] \\
- (\kappa_y + \beta(1-q))\left[\frac{c}{\sigma}\phi_\pi + g\psi_\pi - 1 + p - \frac{c}{\sigma}(1-p)\right]
\end{aligned}
$$

If ψ_π tends to infinity and ψ_y is bounded, the second term on the RHS is positive and, thus, $|\boldsymbol{\mathcal{A}}_{J_1}| < 0$. We have that

$$
\lim_{\psi_\pi \to \infty} |\boldsymbol{\mathcal{A}}_{J_1}| = \lim_{\psi_\pi \to \infty} |\boldsymbol{\mathcal{A}}_{J_3}| = -\infty \tag{87}
$$

We proceed with $|\boldsymbol{\mathcal{A}}_{J_2}| = |\boldsymbol{\mathcal{A}}_{J_4}|$, which is nothing but

$$
\begin{aligned}
|\boldsymbol{\mathcal{A}}_{J_2}| = |\boldsymbol{\mathcal{A}}_{J_4}| = \left[g\psi_y + 1 - p - \frac{c}{\sigma}(1-q)\right](\beta q - 1) \\
- (\kappa_y + \beta(1-q))\left[g\psi_\pi - 1 + p - \frac{c}{\sigma}(1-p)\right]
\end{aligned} \tag{88}
$$

As previously, the second term on the RHS is positive if ψ_π tends to infinity and ψ_y is bounded. Hence, we have that

$$
\lim_{\psi_\pi \to \infty} |\boldsymbol{\mathcal{A}}_{J_2}| = \lim_{\psi_\pi \to \infty} |\boldsymbol{\mathcal{A}}_{J_4}| = -\infty \tag{89}
$$

Thus, we have that if $\psi_\pi \to \infty$, the determinants of $\boldsymbol{\mathcal{A}}_{J_j}$, $j \in \{1, 2, 3, 4\}$, are negative. If $\psi_\pi \to -\infty$, the same logic applies. In this case, the determinants of $\boldsymbol{\mathcal{A}}_{J_j}$, $j \in \{1, 2, 3, 4\}$, are positive. This completes the proof. ■

Lower bound for $\boldsymbol{\psi_\pi}$: We now find the lower bound for ψ_π that guarantees the satisfaction of the CC conditions. For ease of exposition, we assume that $\psi_y = 0$. The model

is coherent and complete, when $|\boldsymbol{\mathcal{A}}_{J_k}| < 0$. Hence ψ_π must satisfy:

$$\begin{cases} \left[1+\frac{c}{\sigma}\phi_y - p - \frac{c}{\sigma}(1-q)\right](\beta q - 1) - (\kappa_y + \beta(1-q))\left[\frac{c}{\sigma}\phi_\pi + g\psi_\pi - 1 + p - \frac{c}{\sigma}(1-p)\right] < 0, \\ \left[1-p-\frac{c}{\sigma}(1-q)\right](\beta q - 1) - (\kappa_y + \beta(1-q))\left[g\psi_\pi - 1 + p - \frac{c}{\sigma}(1-p)\right] < 0. \end{cases}$$

Rearrange to get

$$\begin{cases} \left[1+\frac{c}{\sigma}\phi_y - p - \frac{c}{\sigma}(1-q)\right](\beta q - 1) < (\kappa_y + \beta(1-q))\left[\frac{c}{\sigma}\phi_\pi + g\psi_\pi - 1 + p - \frac{c}{\sigma}(1-p)\right], \\ \left[1-p-\frac{c}{\sigma}(1-q)\right](\beta q - 1) < (\kappa_y + \beta(1-q))\left[g\psi_\pi - 1 + p - \frac{c}{\sigma}(1-p)\right]. \end{cases}$$

Rearrange to get the system that ψ_π must satisfy. Note that depending on the values of monetary policy feedback parameters, ϕ_π and ϕ_y, one of the conditions is redundant

$$\begin{cases} \frac{\left[1+\frac{c}{\sigma}\phi_y - p - \frac{c}{\sigma}(1-q)\right](\beta q-1)}{(\kappa_y+\beta(1-q))} < \frac{c}{\sigma}\phi_\pi + g\psi_\pi - 1 + p - \frac{c}{\sigma}(1-p), \\ \frac{\left[1-p-\frac{c}{\sigma}(1-q)\right](\beta q-1)}{(\kappa_y+\beta(1-q))} < g\psi_\pi - 1 + p - \frac{c}{\sigma}(1-p). \end{cases}$$

If the monetary authority follows strict inflation targeting ($\phi_y = 0$), given the second inequality, the first one is redundant with respect to ψ_π. Thus inspecting the second inequality we have:

$$\left\{\frac{\left[1-p-\frac{c}{\sigma}(1-q)\right](\beta q-1)}{(\kappa_y+\beta(1-q))} + 1 - p + \frac{c}{\sigma}(1-p)\right\}\frac{1}{g} = \underline{\psi_\pi} < \psi_\pi,$$

where $\underline{\psi_\pi}$ denotes the lower bound for ψ_π.

B.4 Canonical Form Coefficients under Consumption Tax Rules

Contemporaneous inflation targeting: $\mathbb{E}_t \Delta \hat{\tau}^c_{t+1} = \psi_\pi \hat{\pi}_t$

$$\boldsymbol{A}_1 = \begin{pmatrix} 1 + \phi_y \sigma^{-1} & \phi_\pi \sigma^{-1} \\ \kappa_y & -1 \end{pmatrix}, \quad \boldsymbol{A}_0 = \begin{pmatrix} 1 & \psi_\pi \Psi^c \sigma^{-1} \\ \kappa_y & -1 \end{pmatrix},$$

$$\boldsymbol{B}_0 = \boldsymbol{B}_1 = \begin{pmatrix} -1 & -\sigma^{-1} \\ 0 & \beta \end{pmatrix}.$$

Contemporaneous inflation and output targeting: $\mathbb{E}_t \Delta \hat{\tau}^c_{t+1} = \psi_\pi \hat{\pi}_t + \psi_y \hat{y}_t$

$$\boldsymbol{A}_1 = \begin{pmatrix} 1 + (\phi_y + \Psi^c \psi_y)\sigma^{-1} & \sigma^{-1}(\phi_\pi + \Psi^c \psi_\pi) \\ \kappa_y & -1 \end{pmatrix}, \quad \boldsymbol{A}_0 = \begin{pmatrix} 1 + \Psi^c \psi_y \sigma^{-1} & \sigma^{-1} \Psi^c \psi_\pi \\ \kappa_y & -1 \end{pmatrix},$$

$$\boldsymbol{B}_1 = \begin{pmatrix} -1 & -\sigma^{-1} \\ 0 & \beta \end{pmatrix}, \quad \boldsymbol{B}_0 = \boldsymbol{B}_1.$$

B.5 NK-FP Model with Inertial Taxes

The canonical form coefficients are given by

$$\boldsymbol{A}_1 = \begin{pmatrix} \kappa_y & -1 & 0 & \kappa_g \\ -1 & 0 & -\frac{c}{\sigma} & g \\ \phi_y & \phi_\pi & -1 & 0 \\ \psi_y & \psi_\pi & 0 & -1 \end{pmatrix}, \quad \boldsymbol{A}_0 = \begin{pmatrix} \kappa_y & -1 & 0 & \kappa_g \\ -1 & 0 & 0 & g \\ \phi_y & \phi_\pi & -1 & 0 \\ \psi_y & \psi_\pi & 0 & -1 \end{pmatrix},$$

$$\boldsymbol{B}_0 = \boldsymbol{B}_1 = \begin{pmatrix} 0 & \beta & 0 & 0 \\ 1 & \frac{c}{\sigma} & 0 & -g \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \boldsymbol{h}_0 = \boldsymbol{h}_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \rho_\tau \end{pmatrix},$$

$$\boldsymbol{C}_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \boldsymbol{C}_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{c\mu}{\sigma} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\boldsymbol{Y}_t = \begin{pmatrix} \hat{x}_t \\ \hat{\pi}_t \\ \hat{i}_t \\ \widehat{GR}_t \end{pmatrix}, \quad \boldsymbol{Y}_{t-1} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \widehat{GR}_{t-1} \end{pmatrix}, \quad \boldsymbol{X}_t = \begin{pmatrix} u_t \\ \varepsilon_t \\ 0 \\ 1 \end{pmatrix}.$$